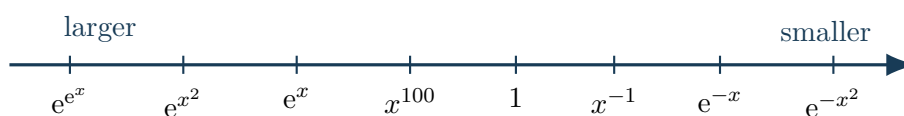


Transseries for Mere Mortals

A Step-by-Step Tutorial from Ordinary Asymptotics
to Exponentials, Logarithms, Stokes Phenomena, and Resurgence



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This tutorial is written for a reader who has completed a standard calculus sequence and has seen elementary differential equations and power series. It synthesizes the materials supplied in the accompanying archive with the foundational literature on logarithmic-exponential transseries, asymptotic differential algebra, Borel summation, and resurgence. The goal is conceptual fluency and practical calculation rather than a bare catalogue of definitions.

Preface: what this book is trying to do

A power series is a remarkable mathematical machine. It can encode a function, solve a differential equation, invert a map, and turn calculus into algebra. Yet it has blind spots. It cannot naturally distinguish two functions whose difference is smaller than every power of the expansion variable. It often produces divergent series near irregular singular points. It can describe a perturbative part of a solution while completely missing free constants carried by exponentially small terms.

A **transseries** is a larger symbolic language designed to repair those blind spots. Its terms may involve powers, logarithms, exponentials, iterated exponentials and logarithms, and infinite sums arranged by asymptotic size. A typical expression might look like

$$x + \log x + \frac{\log x}{x} + Ce^{-x} \left(1 + \frac{2}{x} + \cdots \right) + C^2 e^{-2x} \left(\frac{1}{x} + \cdots \right) + \cdots .$$

In an appropriate formal setting, one can add, multiply, divide, exponentiate, take logarithms, differentiate, integrate, compose, and solve equations with such objects. The resulting field is non-Archimedean: it contains positive infinite elements such as x and positive infinitesimals such as x^{-1} and e^{-x} .

This subject has several overlapping traditions. Écalle’s transseries grew from singular differential equations and resurgence. Dahn and Göring developed related exponential fields from model theory. Hardy fields, Hahn series, computer algebra, surreal numbers, and mathematical physics all meet here. The word *transseries* therefore has variants. We will state conventions rather than pretend there is only one definition.

Big idea

A transseries is best understood as an *asymptotically ordered formal expansion*. The order tells us which term dominates; the support condition makes infinite algebra legal; and closure operations make the language rich enough to follow equations beyond ordinary power series.

The book has five goals:

1. develop intuition for asymptotic scales;
2. construct logarithmic-exponential transseries carefully;
3. teach practical expansion and equation-solving methods;
4. explain divergence, Borel summation, Stokes phenomena, and resurgence;
5. place the subject among Hardy fields, H -fields, surreal numbers, and symbolic computation.

Prerequisites

The main prerequisites are limits, derivatives, integrals, elementary Taylor series, basic complex numbers, and introductory ordinary differential equations. No prior model theory, differential algebra, or surreal-number theory is assumed.

How to read the book

For a first pass, read Chapters 1–4, 8–16, and 19–21. Chapters 5–7 explain support foundations and can be revisited when a formal legality question appears. The later structural chapters form a second course. Exercises are distributed throughout, with a capstone problem set and detailed solutions in the appendices.

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Notation and conventions

Throughout most of the book, x is a formal positive infinite variable, suggesting $x \rightarrow +\infty$. We use

$$f \prec g, \quad f \asymp g, \quad f \sim g$$

for “ f is smaller than g ,” “ f and g have comparable magnitude,” and “ $f/g \rightarrow 1$.” When actual functions are involved,

$$f \prec g \quad \text{corresponds to} \quad f/g \rightarrow 0.$$

We write $\mathbb{T}$ for a field of real logarithmic-exponential transseries and $\mathfrak{G}$ for its multiplicative group of transmonomials. A monomial is denoted $\mathfrak{m}$ or $\mathfrak{n}$.

A formal equality such as

$$(1 - x^{-1})^{-1} = 1 + x^{-1} + x^{-2} + \cdots$$

is equality inside a formal field. An asymptotic relation such as

$$f(x) \sim 1 + x^{-1} + 2x^{-2} + \cdots$$

relates an actual function to its successive truncations. The distinction is essential.

Part I

Why ordinary series are not enough

Asymptotic thinking before transseries

1.1 A limit remembers too little

Suppose $f(x) \rightarrow 1$ as $x \rightarrow +\infty$. This tells us the destination but not the route. The functions

$$1 + \frac{1}{x}, \quad 1 + \frac{1}{\sqrt{x}}, \quad 1 + e^{-x}, \quad 1 + \frac{\sin x}{x}$$

all approach 1, but at very different rates and with different qualitative behavior.

We write $f = O(g)$ when $|f| \leq C|g|$ eventually, and $f = o(g)$ when $f/g \rightarrow 0$. The relation $f \sim g$ means $f/g \rightarrow 1$.

For example,

$$e^{-x} = o(x^{-N}) \quad \text{for every fixed } N > 0.$$

This is proved by repeated l'Hôpital's rule or by the fact that $x^N e^{-x} \rightarrow 0$.

1.2 Asymptotic scales

A sequence (ϕ_n) is an asymptotic scale if

$$\phi_{n+1} \prec \phi_n.$$

The familiar inverse-power scale is

$$1 \succ x^{-1} \succ x^{-2} \succ x^{-3} \succ \dots$$

A function has Poincaré expansion

$$f(x) \sim \sum_{n \geq 0} a_n x^{-n}$$

if for every fixed N ,

$$f(x) - \sum_{n=0}^{N-1} a_n x^{-n} = o(x^{-N+1}).$$

The infinite series need not converge. It is a compact promise about every finite truncation.

Worked example

For $f(x) = 1/(x+1)$,

$$\frac{1}{x+1} = \frac{1}{x} \frac{1}{1+x^{-1}} = x^{-1} - x^{-2} + x^{-3} - \dots$$

Here the series actually converges for $|x| > 1$, but its asymptotic meaning only needs the remainder estimate after each finite truncation.

1.3 Beyond all algebraic orders

The functions 0 and e^{-x} have the same asymptotic expansion in powers of x^{-1} :

$$0 + 0x^{-1} + 0x^{-2} + \dots$$

Indeed, $e^{-x} = o(x^{-N})$ for every N . Thus an inverse-power expansion is not a unique fingerprint.

More generally,

$$f(x) \quad \text{and} \quad f(x) + Ce^{-x}$$

have the same Poincaré expansion for every constant C . The missing constant may be exactly the integration constant of a differential equation.

Big idea

An asymptotic scale can see only differences comparable with one of its members. To distinguish f from $f + e^{-x}$, enlarge the scale so that e^{-x} becomes a visible monomial.

1.4 Divergent series can still be useful

Consider

$$\sum_{n \geq 0} \frac{n!}{x^{n+1}}.$$

For every fixed x , its terms eventually grow, so the series diverges. Yet the first several terms can approximate a function very accurately when x is large. The term ratio is

$$\frac{(n+1)!x^{-n-2}}{n!x^{-n-1}} = \frac{n+1}{x}.$$

Terms decrease until n is about x and then increase. Truncating near the least term produces an exponentially small error. Divergence and exponentials are therefore linked.

1.5 Non-Archimedean arithmetic

In the real numbers, the Archimedean property says that for any positive a, b , some integer multiple of a exceeds b . A field containing an element x larger than every integer violates this property. Its reciprocal x^{-1} is smaller than every positive real constant.

Transseries treat x as genuinely infinite in the formal order:

$$x > n \quad (n \in \mathbb{N}), \quad 0 < x^{-1} < 1/n.$$

This converts asymptotic language into ordered-field arithmetic.

1.6 Three notions that must not be confused

Notion	Question	Example
Formal legality	Is the support compatible with infinite algebra?	$\sum_{n \geq 0} x^{-n}$ is legal as a Hahn series.
Analytic convergence	Does the numerical sum converge for fixed x ?	The same geometric series converges for $x > 1$.
Asymptotic validity	Do truncations approximate a function in a limit?	$\sum n!x^{-n-1}$ may be asymptotic although divergent.

1.7 Why differential equations force the issue

The equation

$$y' + y = \frac{1}{x}$$

has a formal particular solution

$$\frac{1}{x} + \frac{1}{x^2} + \frac{2!}{x^3} + \frac{3!}{x^4} + \cdots,$$

but the general solution also contains Ce^{-x} . A power series alone cannot record C . A transseries can:

$$y = \sum_{n \geq 0} \frac{n!}{x^{n+1}} + Ce^{-x}.$$

This example will return throughout the book.

Exercise 1.1. Prove that $e^{-x^2} \prec e^{-x} \prec x^{-N}$ for every fixed $N > 0$.

Exercise 1.2. Show that x^{-1} and $(x+1)^{-1}$ have the same leading term but different full inverse-power expansions.

Exercise 1.3. Explain why adding $e^{-\sqrt{x}}$ to a function does not change its inverse-power expansion, but does change a scale containing $e^{-\sqrt{x}}$.

A larger alphabet of asymptotic sizes

2.1 The hierarchy

For fixed positive a, b, c , a useful chain is

$$e^{x^2} \succ e^x \succ x^a \succ (\log x)^b \succ 1 \succ (\log x)^{-c} \succ x^{-a} \succ e^{-x} \succ e^{-x^2}.$$

Each neighboring ratio tends to infinity. For instance,

$$\frac{e^x}{x^a} = \exp(x - a \log x) \rightarrow \infty,$$

because $x \gg \log x$.

2.2 Transmonomials

A **transmonomial** is a formal object used as one pure asymptotic size. Examples are

$$x^a, \quad x^a(\log x)^b, \quad e^{x^2-x}, \quad x^{-3}e^{-x}, \quad e^{-e^x}, \quad e^{x/\log x}.$$

Transmonomials form a multiplicative group:

$$mn \in \mathfrak{G}, \quad m^{-1} \in \mathfrak{G}, \quad 1 \in \mathfrak{G}.$$

The order is compatible with multiplication by positive monomials.

2.3 Comparing exponentials

For positive exponentials,

$$e^A \prec e^B \iff A - B \rightarrow -\infty.$$

Thus comparing exponential monomials reduces to comparing exponents.

Worked example

Compare $x^{100}e^{-x}$ and x^{-1000} . Their ratio is

$$\frac{x^{100}e^{-x}}{x^{-1000}} = x^{1100}e^{-x} \rightarrow 0.$$

Hence

$$x^{100}e^{-x} \prec x^{-1000}.$$

No fixed algebraic power can compensate for an exponential loss.

2.4 Nested exponentials and logarithms

The hierarchy has no natural endpoint:

$$e^{e^x} \succ e^{x^2} \succ e^x,$$

and

$$\log x \succ \log \log x \succ \log \log \log x \succ 1$$

when each logarithm is defined and tends to infinity.

The ordinary vocabulary “polynomial versus exponential” is therefore only one level of a much taller scale.

2.5 Coefficients do not determine magnitude

For nonzero real constants a, b and monomials $\mathfrak{m} \succ \mathfrak{n}$, the magnitude of $a\mathfrak{m}$ dominates the magnitude of $b\mathfrak{n}$. In symbols,

$$|a\mathfrak{m}| \succ |b\mathfrak{n}|.$$

A very small fixed coefficient cannot reverse an asymptotic comparison. The coefficient matters only when the monomials are equal or comparable.

2.6 What a transseries looks like

A formal transseries is an ordered sum

$$T = \sum_{\mathfrak{m} \in \text{supp } T} c_{\mathfrak{m}} \mathfrak{m}, \quad c_{\mathfrak{m}} \in \mathbb{R},$$

whose support is arranged from larger monomials to smaller ones and satisfies a well-foundedness condition.

Examples include

$$x + \log x + 3 + \frac{2}{x} - e^{-x} + \frac{e^{-x}}{x} + \dots$$

and

$$\sum_{k \geq 0} C^k e^{-kx} x^{2k} \left(\sum_{n \geq 0} a_{k,n} x^{-n} \right).$$

The second expression is organized into exponential sectors.

2.7 Purely large, constant, and small parts

Every suitable transseries decomposes uniquely as

$$T = L + c + S,$$

where L consists of monomials $\succ 1$, $c \in \mathbb{R}$, and $S \prec 1$ contains monomials below 1. This decomposition is fundamental for exponentiation:

$$e^T = e^L e^c e^S.$$

The first factor is a new monomial, the second a real coefficient, and the third expands as an ordinary power series because S is small.

2.8 Leading term and magnitude

If

$$T = c_0 \mathfrak{m}_0 + c_1 \mathfrak{m}_1 + \cdots, \quad \mathfrak{m}_0 \succ \mathfrak{m}_1 \succ \cdots,$$

then

$$\text{lt}(T) = c_0 \mathfrak{m}_0, \quad \text{mag}(T) = \mathfrak{m}_0.$$

The leading term determines sign, comparison, and often the first move in every calculation.

Common trap

A displayed sequence of terms is not automatically a legal transseries. The support must not climb upward forever or allow infinitely many contributions to one coefficient. Formal support control comes before algebra.

2.9 Changing variables

The same transseries may look different under $g = 1/x$. For example,

$$x^\beta e^{-Ax} = g^{-\beta} e^{-A/g}.$$

Asymptotic differential algebra commonly uses $x \rightarrow +\infty$; physics often uses $g \rightarrow 0^+$. Translation between them is straightforward but changes the visual order of powers.

Exercise 2.1. Order $e^{x/\log x}$, $e^{\sqrt{x}}$, x^{1000} , and $e^{(\log x)^2}$.

Exercise 2.2. Decompose

$$T = e^x + x^3 - 2 + \frac{\log x}{x} + e^{-x}$$

into purely large, constant, and small parts.

Exercise 2.3. Find the leading term of $(x + \log x)^2 - x^2$.

A first guided tour of transseries calculus

This chapter performs the main operations informally before the support machinery is introduced. Each computation follows one principle: factor the dominant term and expand in a small remainder.

3.1 Reciprocal

For

$$T = x + \log x,$$

factor x :

$$T = x \left(1 + \frac{\log x}{x} \right).$$

Since $(\log x)/x \prec 1$,

$$\begin{aligned} T^{-1} &= x^{-1} \left(1 + \frac{\log x}{x} \right)^{-1} \\ &= \frac{1}{x} - \frac{\log x}{x^2} + \frac{(\log x)^2}{x^3} - \frac{(\log x)^3}{x^4} + \dots \end{aligned}$$

3.2 Logarithm

For a positive transseries $T = c\mathbf{m}(1 + U)$ with $c > 0$ and $U \prec 1$,

$$\log T = \log c + \log \mathbf{m} + \left(U - \frac{U^2}{2} + \frac{U^3}{3} - \dots \right).$$

For example,

$$\begin{aligned} \log(e^x + x) &= x + \log(1 + xe^{-x}) \\ &= x + xe^{-x} - \frac{x^2}{2}e^{-2x} + \frac{x^3}{3}e^{-3x} - \dots \end{aligned}$$

3.3 Exponential

If $S \prec 1$, then

$$e^S = 1 + S + \frac{S^2}{2!} + \dots$$

For a general $T = L + c + S$,

$$e^T = \underbrace{e^L}_{\text{monomial coefficient}} \underbrace{e^c}_{\text{monomial coefficient}} \underbrace{\left(1 + S + \frac{S^2}{2} + \cdots\right)}_{\text{small expansion}}.$$

Worked example

Let

$$T = x + 2 + \frac{1}{x} + e^{-x}.$$

Then

$$e^T = e^x e^2 e^{x^{-1} + e^{-x}}.$$

The final factor begins

$$1 + x^{-1} + \frac{1}{2x^2} + e^{-x} + x^{-1}e^{-x} + \frac{1}{2}e^{-2x} + \cdots.$$

The terms must then be sorted by asymptotic size.

3.4 Real powers

Define

$$T^\alpha = \exp(\alpha \log T)$$

for positive T . If $T = x^2(1 + x^{-1})$, then

$$T^\alpha = x^{2\alpha} \left(1 + \alpha x^{-1} + \frac{\alpha(\alpha-1)}{2} x^{-2} + \cdots\right).$$

3.5 Differentiation

Differentiate term by term. A monomial

$$\mathfrak{m} = x^a e^L$$

satisfies

$$\mathfrak{m}' = \mathfrak{m} \left(\frac{a}{x} + L' \right).$$

Thus one input monomial may produce a finite transseries rather than a single monomial. Depending on L' , the leading derivative term can be larger or smaller than the input monomial.

For example,

$$\frac{d}{dx} (x^3 e^{-x^2}) = x^3 e^{-x^2} \left(\frac{3}{x} - 2x \right) = -2x^4 e^{-x^2} + 3x^2 e^{-x^2}.$$

3.6 Integration

Integration is more delicate, but dominant balance often starts it. For

$$\int x^a e^{-x} dx,$$

try

$$-x^a e^{-x} F(x), \quad F = 1 + \frac{a}{x} + \frac{a(a-1)}{x^2} + \dots$$

Differentiation verifies the asymptotic antiderivative. If a is a nonnegative integer, the series terminates.

3.7 Composition

If S is large positive, expressions such as $T(S)$ can often be expanded formally. For

$$T(x) = \log x + x^{-1}, \quad S(x) = x + \log x,$$

we compute

$$T \circ S = \log(x + \log x) + \frac{1}{x + \log x},$$

then expand each piece by factoring x .

3.8 Compositional inversion

To invert

$$T(y) = y + \log y,$$

solve

$$y + \log y = x.$$

Since $y \sim x$, a first correction is $y \approx x - \log x$. Successive substitution gives

$$y = x - \log x + \frac{\log x}{x} + \frac{(\log x)^2 - 2 \log x}{2x^2} + \dots$$

This is $W(e^x)$.

3.9 A mixed exponential inverse

Factor the dominant exponential:

$$\begin{aligned} (e^x + x)^{-1} &= e^{-x} (1 + x e^{-x})^{-1} \\ &= e^{-x} - x e^{-2x} + x^2 e^{-3x} - x^3 e^{-4x} + \dots \end{aligned}$$

This simple example is a model for sector expansions.

3.10 A differential equation preview

For

$$y' + y = \frac{1}{x},$$

substitute $y = \sum a_n x^{-n-1}$. The recurrence is

$$a_0 = 1, \quad a_n = na_{n-1},$$

so $a_n = n!$. The homogeneous solution Ce^{-x} must be added. The formal answer is

$$y = \sum_{n \geq 0} n! x^{-n-1} + Ce^{-x}.$$

Formal differentiation verifies it even though the power-series sector diverges analytically.

Checkpoint

Nearly every elementary transseries calculation follows this pattern:

1. find the dominant monomial;
2. factor it;
3. reduce to a familiar Taylor series in something small;
4. multiply out and reorder;
5. justify the infinite operations using support conditions.

Exercise 3.1. Expand $\log(\cosh x)$ through the e^{-6x} sector.

Exercise 3.2. Expand $(x^2 + x + 1)^{1/2}$ through x^{-4} .

Exercise 3.3. Differentiate $e^{x+\sqrt{x}}x^{-3}$ and identify its leading term.

Part II

The formal engine

Dominance, leading terms, and asymptotic algebra

4.1 Dominance as an order relation

For positive monomials, write

$$\mathbf{m} \succ \mathbf{n}$$

when $\mathbf{m}/\mathbf{n}$ is infinitely large. Equivalently, $\mathbf{n} \prec \mathbf{m}$. We also use

$$\mathbf{m} \asymp \mathbf{n}$$

when their ratio is a nonzero real constant or, for general transseries, tends to one of comparable magnitude.

The order must satisfy:

1. exactly one of $\mathbf{m} \prec \mathbf{n}$, $\mathbf{m} = \mathbf{n}$, or $\mathbf{m} \succ \mathbf{n}$ holds;
2. multiplying by a positive monomial preserves order;
3. inversion reverses order among monomials above 1;
4. exponentiation preserves order of exponents.

4.2 Leading monomial and leading coefficient

For nonzero

$$T = \sum c_{\mathbf{m}} \mathbf{m},$$

let $\mathbf{m}_0$ be the largest support element. Define

$$\text{mag}(T) = \mathbf{m}_0, \quad \text{lc}(T) = c_{\mathbf{m}_0}, \quad \text{lt}(T) = c_{\mathbf{m}_0} \mathbf{m}_0.$$

Then

$$T \sim \text{lt}(T)$$

in the formal asymptotic sense.

4.3 Sign and order of transseries

A nonzero transseries is positive if its leading coefficient is positive. For $S, T \in \mathbb{T}$,

$$S < T \iff T - S > 0.$$

Thus comparison reduces to subtracting and reading the first nonzero term.

Worked example

Compare

$$A = x + \log x + x^{-1}, \quad B = x + 2 \log x - x^{-100}.$$

Their difference is

$$B - A = \log x - x^{-1} - x^{-100}.$$

Its leading term is $\log x > 0$, so $B > A$.

4.4 Dominant equivalence

Write $S \asymp T$ if they have the same magnitude monomial, and $S \sim T$ if their quotient is $1 +$ an infinitesimal. If

$$S = am(1 + U), \quad T = bm(1 + V), \quad U, V \prec 1,$$

then $S \asymp T$, while $S \sim T$ exactly when $a = b$.

4.5 Cancellation

If S and T have different leading monomials, the larger survives in $S + T$. If the leading monomials agree, coefficients may cancel and reveal the next scale.

For example,

$$(x + \log x) - (x + 1) = \log x - 1 \sim \log x.$$

The order of a sum cannot be predicted from the summands' magnitudes alone when cancellation is possible.

4.6 Multiplication

Leading terms multiply:

$$\text{lt}(ST) = \text{lt}(S) \text{lt}(T).$$

Indeed, if $S = am(1 + U)$ and $T = bm(1 + V)$ with $U, V \prec 1$, then

$$ST = abmn(1 + U + V + UV),$$

and the parenthesis is $1 +$ an infinitesimal.

4.7 Valuation language

Some authors encode magnitude by a valuation v satisfying

$$v(ST) = v(S) + v(T), \quad v(S + T) \geq \min\{v(S), v(T)\}.$$

Larger valuation means smaller magnitude. Thus

$$S \prec T \iff v(S) > v(T).$$

This reversal is standard in valuation theory but can be disorienting at first.

4.8 Infinitesimals and infinite elements

An element S is infinitesimal if $S \prec 1$, finite if $S \asymp 1$, and infinite if $S \succ 1$. If S is finite, it has a unique decomposition

$$S = c + \varepsilon, \quad c \in \mathbb{R}, \quad \varepsilon \prec 1.$$

The constant c is its standard part.

4.9 Asymptotic equivalence behaves algebraically

If $S_1 \sim T_1$ and $S_2 \sim T_2$ with nonzero terms, then

$$S_1 S_2 \sim T_1 T_2, \quad \frac{S_1}{S_2} \sim \frac{T_1}{T_2}.$$

But addition requires care: $S_1 + S_2$ may experience cancellation.

Common trap

From $S \sim T$ one may not conclude $e^S \sim e^T$. The ratio is e^{S-T} , which tends to 1 only if $S - T \prec 1$, not merely if $(S - T)/T \prec 1$. For example, $x + 1 \sim x$, but $e^{x+1}/e^x = e$.

4.10 Dominant balance in equations

In a sum equal to zero, a single uniquely dominant nonzero term cannot occur. At least two terms of maximal magnitude must cancel. This elementary observation is the basis of Newton polygons, WKB actions, and sector selection.

For

$$Y^2 - xY - 1 = 0,$$

a branch with $Y \asymp x$ balances Y^2 and xY , while a branch with $Y \asymp x^{-1}$ balances xY and 1.

Exercise 4.1. Compare $x + e^{-x}$ and $x + e^{-\sqrt{x}}$.

Exercise 4.2. Give an example where $S \asymp T$ but $S - T$ is exponentially smaller than both.

Exercise 4.3. Show that if $S \prec 1$, then $(1 + S)^{-1} \sim 1 - S$.

Supports and Hahn series

The support is the hidden skeleton of a formal series. It tells us whether infinite addition and multiplication are meaningful.

5.1 Generalized power series

Let $\mathfrak{G}$ be a totally ordered abelian multiplicative group of monomials. A generalized series is

$$T = \sum_{\mathfrak{m} \in \mathfrak{G}} c_{\mathfrak{m}} \mathfrak{m},$$

with support

$$\text{supp } T = \{\mathfrak{m} : c_{\mathfrak{m}} \neq 0\}.$$

We want to read terms from largest to smaller. Therefore the support should be **reverse well ordered**: every nonempty subset has a largest element.

In additive valuation notation, this is usually stated as a well-ordered support. The difference is only the direction of the monomial order.

5.2 Why reverse well ordering matters

If every nonempty subset has a largest monomial, then:

- every nonzero series has a leading term;
- there is no endless climb toward larger and larger support elements;
- recursive calculations can proceed from the top downward.

The support

$$\{1, x^{-1}, x^{-2}, \dots\}$$

is reverse well ordered. The support

$$\{x, x^2, x^3, \dots\}$$

is not, because it has no largest element.

5.3 A subtle example

The set

$$\{x^{1/n} : n \geq 1\}$$

is reverse well ordered under dominance at $x \rightarrow \infty$: every subset of positive integers has a least index, hence a largest monomial. Yet the monomials approach 1 from above. Such a support is legal in a broad Hahn setting but not contained in many finitely generated grids.

By contrast,

$$\{e^{-x/n} : n \geq 1\}$$

has no largest element, because $e^{-x/(n+1)} \succ e^{-x/n}$ for every n .

5.4 Addition

Addition is coefficientwise:

$$\left(\sum c_{\mathbf{m}} \mathbf{m}\right) + \left(\sum d_{\mathbf{m}} \mathbf{m}\right) = \sum (c_{\mathbf{m}} + d_{\mathbf{m}}) \mathbf{m}.$$

The union of two reverse well-ordered supports is reverse well ordered.

5.5 Multiplication

Multiplication is the Cauchy product:

$$ST = \sum_{\rho \in \mathfrak{G}} \left(\sum_{\mathbf{m}\mathbf{n}=\rho} c_{\mathbf{m}} d_{\mathbf{n}} \right) \rho.$$

Two facts are required:

1. the product support must remain reverse well ordered;
2. for each ρ , only finitely many pairs $(\mathbf{m}, \mathbf{n})$ may contribute a nonzero product.

A classical Hahn-series theorem guarantees these properties for well-ordered supports in an ordered abelian group.

5.6 Summable families

An indexed family $(T_i)_{i \in I}$ is formally summable if:

1. the union of all supports is reverse well ordered;
2. each monomial occurs in only finitely many T_i .

Then

$$\sum_{i \in I} T_i$$

is defined coefficientwise.

Worked example

The family $T_n = x^{-n}$ is summable: the union support is reverse well ordered and each monomial occurs once. The family

$$T_n = 1 + x^{-n}$$

is not summable, because the monomial 1 occurs in infinitely many members, leading to an undefined infinite coefficient.

5.7 Formal convergence

A sequence S_N converges formally to S if, for every monomial cutoff, the coefficients above that cutoff eventually stabilize. This is not numerical convergence. For the geometric partial sums

$$S_N = 1 + x^{-1} + \cdots + x^{-N},$$

each coefficient eventually stabilizes, so

$$S_N \longrightarrow \sum_{n \geq 0} x^{-n}$$

formally.

5.8 Neumann series

If $U \prec 1$ and the powers (U^n) form a summable family, then

$$(1 - U)^{-1} = \sum_{n \geq 0} U^n.$$

The proof is formal:

$$(1 - U) \sum_{n=0}^N U^n = 1 - U^{N+1},$$

and U^{N+1} moves below every fixed cutoff as N grows.

5.9 Why analytic convergence is independent

The formal series

$$\sum_{n \geq 0} n! x^{-n}$$

has the same support as a geometric series and is perfectly legal as a Hahn series, though it diverges numerically for every fixed x . Conversely, a numerically convergent series may have a support inappropriate for a chosen formal field.

5.10 Hahn fields

The collection of generalized series with coefficients in a field k and well-ordered support in an ordered abelian group G is denoted in additive notation by

$$k((G)).$$

It is a field. Transseries enrich a Hahn-field architecture by building a sophisticated monomial group recursively using exponentials and logarithms.

Optional advanced viewpoint

Hahn's theorem is foundational because it separates two jobs: the ordered group controls monomial sizes, while well-ordered support controls infinite algebra. Transseries complicate the monomial group but retain this architecture.

Exercise 5.1. Decide whether $\{x^{-n^2} : n \geq 0\}$ is reverse well ordered.

Exercise 5.2. Explain why the family $T_n = x^{-n} + x^{-n-1}$ is summable.

Exercise 5.3. Give a family whose union support is reverse well ordered but that fails point-finiteness.

Grids, ratio sets, and formal convergence

Full well-based supports are flexible, but explicit calculation often needs finite combinatorial data. Grid-based transseries provide it.

6.1 Small ratios

Choose finitely many monomials

$$\boldsymbol{\mu} = \{\mu_1, \dots, \mu_n\}, \quad \mu_i \prec 1.$$

For an integer vector $\mathbf{k} = (k_1, \dots, k_n)$, write

$$\boldsymbol{\mu}^{\mathbf{k}} = \mu_1^{k_1} \cdots \mu_n^{k_n}.$$

A grid is a set

$$\mathfrak{m}_0 \boldsymbol{\mu}^{\mathbf{k}}, \quad \mathbf{k} \geq \mathbf{k}_0$$

coordinatewise. Moving upward in any exponent multiplies by a small ratio and therefore moves downward in magnitude.

6.2 Example of a grid

Let

$$\mu_1 = x^{-1}, \quad \mu_2 = e^{-x}.$$

Then a grid contains monomials

$$x^{-m} e^{-nx}, \quad m \geq m_0, \quad n \geq n_0.$$

A transseries such as

$$1 + \frac{1}{x} + \frac{2}{x^2} + e^{-x} \left(x^2 + 3x + \frac{4}{x} + \cdots \right) + e^{-2x} (1 + \cdots)$$

lives in a shifted grid generated by these ratios, after allowing a suitable leading monomial.

6.3 Ratio sets as bookkeeping certificates

A finite ratio set can certify:

- that one monomial is smaller than another;
- that a series is supported in a grid;
- that powers of a small transseries move downward;
- that a proposed infinite sum is formally convergent.

Edgar calls refined versions of such certificates **witnesses** [16].

6.4 Why finite generation helps

If the support is encoded by integer vectors, multiplication adds vectors:

$$\mu^k \mu^l = \mu^{k+l}.$$

This makes support propagation algorithmic. Truncation becomes a search over a finite region above a cutoff.

6.5 A grid-based geometric series

If

$$U = x^{-1} + e^{-x},$$

then powers of U involve

$$x^{-j} e^{-kx}, \quad j, k \geq 0,$$

with $j + k = n$ inside U^n . The reciprocal

$$(1 - U)^{-1} = 1 + U + U^2 + \dots$$

therefore lies in the grid generated by x^{-1} and e^{-x} .

6.6 A support not contained in a finite grid

The monomials

$$\{x^{1/n} : n \geq 1\}$$

form a reverse-well-ordered support: every nonempty subset of positive integer indices has a least index and hence a largest monomial. Their exponents have unbounded denominators, so no finite set of fixed rational-power ratios generates all of them by integer products. This support is well based but not grid based.

6.7 Formal truncation

Choose a cutoff monomial η . To compute T modulo terms smaller than η , retain only

$$\{\mathfrak{m} \in \text{supp } T : \mathfrak{m} \succeq \eta\}.$$

For a grid, this often corresponds to finitely many exponent vectors. A trustworthy computation should always state its cutoff.

6.8 Geometric convergence in a grid

If $U \prec 1$ is grid based, powers U^n eventually fall below any fixed grid cutoff. Therefore analytic-looking formulas such as

$$\log(1 + U) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n$$

are formally legitimate.

6.9 Witnessing inequalities

Suppose

$$A = x^{-2} + e^{-x}, \quad B = x^{-1}.$$

Both ratios

$$\frac{x^{-2}}{x^{-1}} = x^{-1}, \quad \frac{e^{-x}}{x^{-1}} = xe^{-x}$$

are small. The ratio set

$$\{x^{-1}, xe^{-x}\}$$

certifies $A \prec B$ term by term.

Common trap

A grid is not merely a rectangular picture of the first displayed terms. It is a finite-generation statement about every monomial that can appear. Hidden operations may require enlarging the ratio set.

6.10 Grid-based versus well-based practice

Grid-based fields are excellent for explicit algorithms and many naturally occurring equations. Well-based fields are conceptually broader and may have stronger closure. One often proves structural theorems in the well-based world and performs calculations in a grid-based subfield.

Exercise 6.1. Find a ratio set for the support of

$$x^3 \sum_{m,n \geq 0} 2^{-m-n} x^{-m} e^{-nx}.$$

Exercise 6.2. Explain why $\{x^{1/n} : n \geq 1\}$ is not generated by finitely many rational powers of x .

Exercise 6.3. For $U = x^{-1} + xe^{-x}$, list all monomials in U^3 .

Constructing logarithmic-exponential transseries

We now resolve an apparent circularity: monomials involve exponentials of transseries, while transseries are sums of monomials. The construction proceeds in stages.

7.1 Height zero

Begin with power monomials

$$\mathfrak{G}_0 = \{x^a : a \in \mathbb{R}\}.$$

The corresponding Hahn field consists of generalized power series

$$\mathbb{T}_0 = \mathbb{R}((\mathfrak{G}_0)).$$

Grid-based versions restrict supports to finitely generated exponent grids.

7.2 Purely large exponents

Let $L \in \mathbb{T}_0$ be purely large, meaning every monomial in L is above 1. Adjoin a monomial

$$e^L.$$

Two such exponentials compare by their exponents:

$$e^{L_1} \prec e^{L_2} \iff L_1 - L_2 < 0 \text{ with infinite magnitude at leading order.}$$

7.3 Height one

The height-one monomial group contains products

$$x^a e^L, \quad L \in \mathbb{T}_0 \text{ purely large.}$$

Taking generalized series over this group produces $\mathbb{T}_1$.

Examples include

$$e^{x^2+x}, \quad x^{-3}e^{\sqrt{x}}, \quad e^{-x+\sqrt{x}}, \quad e^{x^{-1}} \text{ after small expansion rather than as a new large monomial.}$$

7.4 Why constants and small terms stay outside the monomial exponent

If

$$T = L + c + S, \quad S \prec 1,$$

then

$$e^T = e^L e^c \left(1 + S + \frac{S^2}{2} + \cdots \right).$$

The large part creates the monomial, the constant becomes its coefficient, and the small part becomes a series. This normalization avoids redundant representations.

7.5 Higher exponential height

Repeat the process:

$$\mathbb{T}_0 \subset \mathbb{T}_1 \subset \mathbb{T}_2 \subset \cdots.$$

At height two, exponents may contain height-one monomials, so e^{e^x} appears. The union over all finite heights gives a log-free exponential transseries field.

7.6 Adding logarithmic depth

To include $\log x$, substitute a logarithmically smaller infinite variable into the log-free construction. Informally, allow

$$T(\log x), \quad T(\log \log x), \quad \dots$$

for finitely many nested logarithms. This introduces monomials such as

$$(\log x)^a, \quad e^{x/\log x}, \quad \frac{x}{\log x}, \quad e^{\sqrt{\log x}}.$$

Example 7.1. The expression

$$\frac{x}{\log x} + \frac{x \log \log x}{(\log x)^2}$$

has finite logarithmic depth and belongs naturally to the full logarithmic-exponential field.

7.7 The apparent circularity disappears

The staged construction is:

1. start with real powers of x ;
2. form generalized series with controlled support;
3. exponentiate only purely large elements to create new monomials;
4. form generalized series over the enlarged group;
5. iterate exponential height finitely many times;

6. allow finite logarithmic substitutions.

At each stage the ingredients already exist before the next stage is defined.

7.8 Normal form

A typical monomial is written

$$\mathfrak{m} = x^a e^L$$

with L purely large and normalized. A typical positive transseries is

$$T = c\mathfrak{m}(1 + U), \quad c > 0, \quad U \prec 1.$$

This normal form makes logarithms, powers, and comparison canonical.

7.9 Closure properties

A standard real LE-transseries field is designed to be closed under:

$$+, -, \times, ^{-1}, \exp, \log, \frac{d}{dx}, \text{integration, and suitable composition.}$$

The exact theorem depends on whether one uses grid-based or well-based transseries and on the chosen definition of composition. Foundational treatments include van der Hoeven, Edgar, and van den Dries–Macintyre–Marker [10, 13, 22].

7.10 Finite height does not mean finite length

A transseries may have infinitely many terms while using only one exponential height. For example,

$$\sum_{n \geq 0} x^n e^{-(n+1)x}$$

has infinitely many sectors but no nested exponential. Conversely, e^{e^x} is a single term of greater height.

7.11 Uniqueness of representation

Normal-form conventions ensure that the same object is not represented both as a monomial and as an expanded small exponential. For example,

$$e^{x^{-1}}$$

is represented as

$$1 + x^{-1} + \frac{1}{2}x^{-2} + \cdots,$$

because its exponent is small, whereas e^x is a monomial because its exponent is large.

Exercise 7.1. Classify the exponential height and logarithmic depth of $e^{e^{\sqrt{\log x}}}$ under a reasonable convention.

Exercise 7.2. Decompose $T = x^2 + 3 + e^{-x}$ into $L + c + S$ and write e^T in normalized form.

Exercise 7.3. Explain why treating $e^{x^{-1}}$ as an independent monomial would destroy uniqueness.

Field arithmetic in slow motion

8.1 Addition and subtraction

Merge supports and combine coefficients. The only conceptual danger is cancellation of leading terms.

Worked example

Let

$$A = x + 2 \log x + 1 + x^{-1}, \quad B = -x - 2 \log x + 3 - x^{-2}.$$

Then

$$A + B = 4 + x^{-1} - x^{-2}.$$

The leading scale drops from x to 1 because two successive levels cancel.

8.2 Multiplication

Multiply distributively and collect monomials. If

$$A = x + 1 + x^{-1}, \quad B = x - 1 + x^{-1},$$

then

$$AB = x^2 + 1 + x^{-2}.$$

Several intermediate terms cancel.

8.3 Reciprocal by normalization

For nonzero T , write

$$T = c\mathfrak{m}(1 + U), \quad U \prec 1.$$

Then

$$T^{-1} = c^{-1}\mathfrak{m}^{-1}(1 - U + U^2 - U^3 + \cdots).$$

This proves every nonzero transseries has a reciprocal, provided the powers of U are summable.

Worked example

For

$$T = x^2 + x + \log x,$$

write

$$T = x^2 \left(1 + x^{-1} + x^{-2} \log x \right).$$

Let $U = x^{-1} + x^{-2} \log x$. Through x^{-5} ,

$$\begin{aligned} T^{-1} &= x^{-2} - x^{-3} + (1 - \log x)x^{-4} \\ &\quad + (2 \log x - 1)x^{-5} + \dots \end{aligned}$$

8.4 Division

Compute $A/B = A \cdot B^{-1}$. The leading term is

$$\text{lt}(A/B) = \frac{\text{lt}(A)}{\text{lt}(B)}.$$

8.5 Solving by coefficient matching

To solve

$$(1 - x^{-1})Y = 1,$$

let $Y = \sum_{n \geq 0} a_n x^{-n}$. Matching gives $a_0 = 1$ and $a_n = a_{n-1}$, so

$$Y = 1 + x^{-1} + x^{-2} + \dots$$

This is equivalent to the Neumann-series formula.

8.6 Formal geometric and binomial identities

For $U \prec 1$,

$$(1 - U)^{-1} = \sum_{n \geq 0} U^n,$$

$$(1 + U)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} U^n,$$

$$\log(1 + U) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n,$$

$$e^U = \sum_{n \geq 0} \frac{U^n}{n!}.$$

They are formal identities because powers of a small U move downward in support.

8.7 Precision bookkeeping

If the desired answer is modulo terms $\prec \eta$, do not compute every intermediate expansion to the same naive number of terms. Multiplication by a large outer monomial can lift a small inner term above the final cutoff.

For example, to compute

$$e^x e^{x^{-1}}$$

through $x^{-3}e^x$, one needs $e^{x^{-1}}$ through x^{-3} , not merely through an absolute cutoff such as x^{-3} disconnected from the outer factor.

8.8 Formal field laws

Because Hahn addition and multiplication are defined coefficientwise with support control, the usual field laws hold. Infinite sums are never rearranged arbitrarily; admissibility guarantees that each coefficient is determined by finitely many contributions.

Exercise 8.1. Compute $(x + 1 + x^{-1})^{-1}$ through x^{-5} .

Exercise 8.2. Expand $(1 + x^{-1} + e^{-x})^{-2}$ through algebraic order x^{-3} and the first exponential sector.

Exercise 8.3. Find the leading term of

$$\frac{e^x + x}{e^x - x} - 1.$$

Exponential, logarithm, and real powers

9.1 Exponentiating a general transseries

Decompose

$$T = L + c + S, \quad L \succ 1 \text{ termwise}, \quad c \in \mathbb{R}, \quad S \prec 1.$$

Then define

$$e^T = e^L e^c \sum_{n \geq 0} \frac{S^n}{n!}.$$

This is both a definition and a normal-form algorithm.

9.2 Exponent laws

The construction is arranged so that

$$e^{S+T} = e^S e^T, \quad e^0 = 1,$$

and $e^T > 0$. Formal proof requires checking summability of the small exponential factors.

9.3 Logarithm of a positive transseries

For positive T , write

$$T = c\mathfrak{m}(1 + U), \quad c > 0, \quad U \prec 1.$$

Then

$$\log T = \log c + \log \mathfrak{m} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n.$$

If $\mathfrak{m} = x^a e^L$, then

$$\log \mathfrak{m} = a \log x + L.$$

Worked example

For

$$T = 3x^2 e^{\sqrt{x}} (1 + x^{-1} + e^{-x}),$$

we obtain

$$\begin{aligned} \log T = & \log 3 + 2 \log x + \sqrt{x} \\ & + x^{-1} - \frac{1}{2}x^{-2} + \frac{1}{3}x^{-3} + e^{-x} - x^{-1}e^{-x} + \dots \end{aligned}$$

9.4 Inverse relationship

For suitable transseries,

$$\log(e^T) = T, \quad e^{\log T} = T.$$

The normal-form decomposition is crucial: it tells the logarithm how to recover large exponent, coefficient, and small unit.

9.5 Real powers

For $T > 0$ and $\alpha \in \mathbb{R}$, define

$$T^\alpha = e^{\alpha \log T}.$$

This agrees with integer powers and gives

$$T^{\alpha+\beta} = T^\alpha T^\beta.$$

9.6 Powers with transseries exponents

One may also define

$$T^S = e^{S \log T}$$

when $T > 0$. This can create new exponential height. For example,

$$x^x = e^{x \log x}.$$

9.7 Expansion around a dominant term

For $A > 0$ and $R \prec A$,

$$\log(A + R) = \log A + \log \left(1 + \frac{R}{A}\right),$$

and

$$(A + R)^\alpha = A^\alpha \left(1 + \frac{R}{A}\right)^\alpha.$$

The same dominant-factor method works at every transseries scale.

9.8 The logarithm of hyperbolic functions

For large x ,

$$\sinh x = \frac{e^x}{2}(1 - e^{-2x}),$$

so

$$\log \sinh x = x - \log 2 - \sum_{n \geq 1} \frac{e^{-2nx}}{n}.$$

This is a genuine exponential transseries invisible to inverse powers.

9.9 Nested operations

To expand

$$\log \log(e^x + x),$$

first expand the inner logarithm, then factor its dominant term x before taking the outer logarithm. Expanding in the wrong order can hide the small parameter.

Common trap

The rule $\log(1 + U) = U - U^2/2 + \dots$ requires $U \prec 1$. It is invalid to apply it directly to $1 + x$ at $x \rightarrow \infty$. First factor the dominant x .

9.10 Complex branches

For complex transseries, logarithms require branch choices, and exponentials may no longer be ordered by positivity. The formal algebra remains useful, but Stokes sectors and phase information replace a single real total order.

Exercise 9.1. Expand $\log(x^2 + x + 1)$ through x^{-4} .

Exercise 9.2. Expand $(e^x + x)^{1/2}$ through the e^{-3x} sector.

Exercise 9.3. Compute the first three terms of $x^{1+1/x}$ relative to its leading monomial.

Differentiation and integration

10.1 Derivative of monomials

For

$$\mathfrak{m} = x^a e^L,$$

we have

$$\mathfrak{m}' = \mathfrak{m} \left(\frac{a}{x} + L' \right).$$

The logarithmic derivative

$$\mathfrak{m}^\dagger = \frac{\mathfrak{m}'}{\mathfrak{m}}$$

is therefore $a/x + L'$.

10.2 Termwise differentiation

Define

$$\left(\sum c_{\mathfrak{m}} \mathfrak{m} \right)' = \sum c_{\mathfrak{m}} \mathfrak{m}'.$$

A theorem is needed to show that the derivative family is summable and has admissible support. This holds in standard LE-transseries fields.

10.3 Examples

$$\begin{aligned} (x^a)' &= ax^{a-1}, \\ (\log x)^{b'} &= \frac{b(\log x)^{b-1}}{x}, \end{aligned}$$

and

$$(x^a e^{-x^2})' = -2x^{a+1} e^{-x^2} + ax^{a-1} e^{-x^2}.$$

The derivative may have a larger polynomial factor while remaining on the same exponential scale.

10.4 Chain rule

For composable transseries,

$$(T \circ S)' = (T' \circ S)S'.$$

This is essential for inverse functions and iteration. Its proof is formal but relies on a legal definition of composition.

10.5 Asymptotic integration of exponential monomials

Suppose

$$f = e^L x^a$$

and L' dominates a/x . A first antiderivative approximation is

$$g_0 = \frac{f}{L'}.$$

Indeed,

$$g'_0 = f + \text{smaller correction}$$

under suitable hypotheses. Correct the residual recursively.

For $f = e^{-x} x^a$, $L' = -1$, so start with $-x^a e^{-x}$. Repeated correction gives

$$\int x^a e^{-x} dx \sim -e^{-x} x^a \left(1 + \frac{a}{x} + \frac{a(a-1)}{x^2} + \cdots \right).$$

10.6 Integration of algebraic monomials

For $a \neq -1$,

$$\int x^a dx = \frac{x^{a+1}}{a+1}.$$

At the resonant exponent $a = -1$,

$$\int x^{-1} dx = \log x.$$

This is one reason logarithmic closure is necessary.

Likewise,

$$\int \frac{dx}{x \log x} = \log \log x,$$

forcing successive logarithmic depths.

10.7 Integration by parts as a transseries algorithm

For

$$I(x) = \int^x e^{P(t)} Q(t) dt,$$

choose $R_0 = Q/P'$ and write

$$(e^P R_0)' = e^P (Q + R'_0).$$

Then subtract an antiderivative of the smaller residual $e^P R'_0$. Iteration produces an asymptotic series.

10.8 Constants of integration

Formal differentiation kills real constants, so antiderivatives are determined only up to $C \in \mathbb{R}$. In differential equations, additional constants may multiply exponentially small homogeneous solutions rather than appear as additive constants in the original variable.

10.9 Integration may enlarge the field

Not every elementary-looking integrand has an elementary antiderivative, but the transseries field often contains a formal antiderivative. Exact analytic representation may involve special functions such as Ei, erf, or incomplete gamma functions.

10.10 Differential field properties

The derivation satisfies

$$(S + T)' = S' + T', \quad (ST)' = S'T + ST',$$

and has constant field $\mathbb{R}$ in the standard real transseries setting. These properties make $\mathbb{T}$ an ordered differential field.

Exercise 10.1. Find an asymptotic antiderivative of $x^{-2}e^{-x}$ through three correction terms.

Exercise 10.2. Differentiate $e^{x/\log x}$ and identify the leading logarithmic factor.

Exercise 10.3. Explain why repeated integration of $1/[x \log x \log \log x]$ forces another logarithmic level.

Composition, inversion, and fixed points

11.1 When composition makes sense

For $T \circ S$, the inner transseries S should normally be large positive so that powers and logarithms in T can be evaluated asymptotically. In addition, substituting S into every monomial of an infinite support must produce a summable family.

11.2 Composition of basic monomials

If $S \succ 1$, then

$$x^a \circ S = S^a, \quad e^L \circ S = e^{L \circ S}, \quad \log x \circ S = \log S.$$

Extend linearly to a summable support.

Worked example

Let

$$T(x) = \log x + x^{-1}, \quad S = x + \log x.$$

Then

$$\begin{aligned} \log S &= \log x + \frac{\log x}{x} - \frac{(\log x)^2}{2x^2} + \cdots, \\ S^{-1} &= x^{-1} - \frac{\log x}{x^2} + \frac{(\log x)^2}{x^3} + \cdots. \end{aligned}$$

Adding gives $T \circ S$.

11.3 Taylor expansion under composition

If $S = x + H$ with $H \prec x$, then formally

$$T(x + H) = \sum_{n \geq 0} \frac{T^{(n)}(x)}{n!} H^n$$

when the family is summable. This is useful for near-identity maps, but the correct smallness condition is relative to the variation scale of T , not merely $H \prec x$.

For $T = e^x$, even $H = 1$ changes the result by a factor e ; one should not call that a small relative correction.

11.4 Compositional inverse

A transseries T with $T \succ 1$ and appropriate monotonicity often has an inverse S satisfying

$$T \circ S = x, \quad S \circ T = x.$$

Start from the dominant inverse and correct recursively.

Procedure

To invert T asymptotically:

1. invert the dominant monomial or leading map;
2. substitute the approximation and compute the residual;
3. divide the residual by the derivative of the map;
4. add the negative correction;
5. repeat at successively smaller scales.

11.5 Inverse of $x + \log x$

The inverse S satisfies

$$S + \log S = x.$$

Since $S \sim x$, begin with $S_0 = x - \log x$. Continuing gives

$$\begin{aligned} S(x) = x - \log x + \frac{\log x}{x} + \frac{(\log x)^2 - 2 \log x}{2x^2} \\ + \frac{2(\log x)^3 - 9(\log x)^2 + 6 \log x}{6x^3} + \dots \end{aligned}$$

This is $W(e^x)$.

11.6 Fixed-point equations

Many equations can be rearranged as

$$Y = \Phi(Y).$$

If

$$\Phi(Y) - \Phi(Z) \prec Y - Z$$

in the relevant asymptotic class, repeated substitution is a formal contraction.

For

$$Y = x + Y^{-1},$$

we have

$$\Phi(Y) - \Phi(Z) = -\frac{Y - Z}{YZ}.$$

When $Y, Z \sim x$, the factor $(YZ)^{-1} \sim x^{-2}$ shrinks the error.

11.7 Newton iteration

For $F(Y) = 0$, formal Newton iteration is

$$Y_{n+1} = Y_n - \frac{F(Y_n)}{F'(Y_n)}.$$

With a correct dominant balance and nonzero leading derivative, each step often improves many orders at once.

11.8 A glimpse of fractional iteration

Integer iterates are

$$T^{[n]} = \underbrace{T \circ \dots \circ T}_{n \text{ times}}.$$

A fractional iterate seeks $T^{[s]}$ satisfying

$$T^{[s]} \circ T^{[t]} = T^{[s+t]}.$$

If V solves Abel's equation

$$V \circ T = V + 1,$$

then

$$T^{[s]} = V^{[-1]} \circ (V + s).$$

Transseries provide a natural language for constructing V and its inverse in many cases [14, 15].

11.9 Composition is not automatic

Even if every individual $\mathfrak{m} \circ S$ is defined, the infinite family may fail summability. Rigorous composition theorems track supports, ratio sets, and the large positive behavior of S [14].

Exercise 11.1. Use Taylor expansion to compute $\sqrt{x^2 + x + 1}$ through order x^{-3} .

Exercise 11.2. Find the first three corrections to the inverse of $T(x) = x + x^{-1}$.

Exercise 11.3. Show that $V(x) = \log x / \log 2$ satisfies $V(2x) = V(x) + 1$ and deduce the fractional iterates of $T(x) = 2x$.

Part III

A practical workshop: building and solving with transseries

The expansion laboratory

This chapter turns the formal rules into a repeatable method.

Big idea

Factor out the dominant monomial. What remains should be $1 +$ something small. Then use an ordinary Taylor series in that small remainder.

12.1 The universal pattern

If $R \prec A \neq 0$, write

$$A + R = A(1 + U), \quad U = R/A \prec 1.$$

Then

$$\begin{aligned} (A + R)^{-1} &= A^{-1} \sum_{n \geq 0} (-U)^n, \\ \log(A + R) &= \log A + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n, \\ (A + R)^\alpha &= A^\alpha \sum_{n \geq 0} \binom{\alpha}{n} U^n. \end{aligned}$$

For exponentials, split $T = L + c + S$ and use $e^T = e^L e^c e^S$.

Procedure

1. Find the largest monomial in every sum.
2. Factor it out.
3. Name the remaining small quantity U .
4. Apply the familiar Taylor series.
5. Order all resulting monomials.
6. Check that every omitted term lies below the requested cutoff.

12.2 Mixed-scale reciprocal

Worked example

Expand

$$\frac{1}{e^x + x + 1}.$$

Factor e^x :

$$\frac{1}{e^x + x + 1} = e^{-x} (1 + (x + 1)e^{-x})^{-1}.$$

Hence

$$\frac{1}{e^x + x + 1} = e^{-x} - (x + 1)e^{-2x} + (x + 1)^2 e^{-3x} - (x + 1)^3 e^{-4x} + \dots$$

Within each exponential sector, expand the polynomial only as far as needed.

12.3 Exponentially accurate logarithms

Since

$$\sinh x = \frac{e^x}{2}(1 - e^{-2x}),$$

we get

$$\log(\sinh x) = x - \log 2 - e^{-2x} - \frac{1}{2}e^{-4x} - \frac{1}{3}e^{-6x} - \dots$$

Similarly,

$$\log(\cosh x) = x - \log 2 + e^{-2x} - \frac{1}{2}e^{-4x} + \frac{1}{3}e^{-6x} - \dots$$

Subtracting gives

$$\log(\coth x) = 2e^{-2x} + \frac{2}{3}e^{-6x} + \frac{2}{5}e^{-10x} + \dots$$

12.4 Nested logarithms

Worked example

First,

$$\log(e^x + x) = x + xe^{-x} - \frac{x^2}{2}e^{-2x} + \frac{x^3}{3}e^{-3x} + \dots$$

Factor x before applying the outer logarithm:

$$\log \log(e^x + x) = \log x + \log \left(1 + e^{-x} - \frac{x}{2}e^{-2x} + \frac{x^2}{3}e^{-3x} + \dots \right).$$

Expanding gives

$$\begin{aligned} \log \log(e^x + x) &= \log x + e^{-x} - \frac{x+1}{2}e^{-2x} \\ &\quad + \left(\frac{x^2}{3} + \frac{x}{2} + \frac{1}{3}\right)e^{-3x} + \dots \end{aligned}$$

12.5 Exponentiating several small scales

Let

$$S = x^{-1} + 2x^{-2} + e^{-x}.$$

Then

$$e^S = e^{x^{-1}+2x^{-2}}e^{e^{-x}}.$$

The algebraic factor is

$$1 + x^{-1} + \frac{5}{2}x^{-2} + \frac{13}{6}x^{-3} + \dots,$$

and the exponential-sector factor is

$$1 + e^{-x} + \frac{1}{2}e^{-2x} + \dots.$$

Multiplication creates mixed terms such as $x^{-1}e^{-x}$.

12.6 Cutoffs

Choose a cutoff monomial η and write

$$T \equiv S \pmod{o(\eta)}$$

when $T - S \prec \eta$. This is more informative than saying “plus higher-order terms” in a multiscale expansion.

12.7 Cancellation

For

$$\sqrt{x^2 + 1} - x,$$

the two displayed terms have size x , but cancellation reveals

$$\sqrt{x^2 + 1} - x = \frac{1}{2x} - \frac{1}{8x^3} + \frac{1}{16x^5} - \dots.$$

Rationalization or a sufficiently deep expansion prevents loss of the hidden scale.

Exercise 12.1. Expand $(e^x + x^2)^{-2}$ through the e^{-4x} sector.

Exercise 12.2. Find the first four nonzero exponential terms in $\log(1 + \tanh x)$.

Exercise 12.3. Order x^5e^{-2x} , $x^{-100}e^{-x}$, $e^{-x-\sqrt{x}}$, $e^{-x}/\log x$, and $e^{-x-x^{-1}}$.

Algebraic and implicit equations

13.1 Dominant balance

If a sum equals zero, at least two dominant terms must cancel. In

$$Y^2 - xY - 1 = 0,$$

one branch has $Y \asymp x$ and another has $Y \asymp x^{-1}$.

The exact roots are

$$Y_{\pm} = \frac{x \pm \sqrt{x^2 + 4}}{2},$$

so

$$\begin{aligned} Y_+ &= x + x^{-1} - x^{-3} + 2x^{-5} - 5x^{-7} + \cdots, \\ Y_- &= -x^{-1} + x^{-3} - 2x^{-5} + 5x^{-7} - \cdots. \end{aligned}$$

13.2 Residual correction

For $F(Y) = 0$, let $R = F(Y_0)$. Writing $Y = Y_0 + \delta$ gives

$$0 = R + F'(Y_0)\delta + \text{smaller terms},$$

so

$$\delta \sim -\frac{R}{F'(Y_0)}.$$

This is asymptotic Newton iteration.

13.3 An implicit logarithmic equation

Consider

$$Y = x + \log Y + Y^{-1}.$$

Let $L = \log x$ and seek

$$Y = x + L + \frac{A(L)}{x} + \frac{B(L)}{x^2} + \cdots.$$

Expanding the right side gives

$$A = L + 1, \quad B = 1 - \frac{L^2}{2},$$

so

$$Y = x + L + \frac{L+1}{x} + \frac{1-L^2/2}{x^2} + \dots.$$

The coefficient at each inverse power is a polynomial in $\log x$.

13.4 Formal contraction

For

$$Y = x + Y^{-1},$$

set $\Phi(Y) = x + Y^{-1}$. Then

$$\Phi(Y) - \Phi(Z) = -\frac{Y - Z}{YZ}.$$

On $Y, Z \sim x$, the difference shrinks by x^{-2} , so iteration determines successive coefficients.

13.5 Newton polygon intuition

For

$$Y^3 + xY + x^4 = 0,$$

try $Y \asymp x^\alpha$. The term exponents are

$$3\alpha, \quad 1 + \alpha, \quad 4.$$

The only consistent top balance is $3\alpha = 4$, hence $\alpha = 4/3$ and

$$Y \sim \omega x^{4/3}, \quad \omega^3 = -1.$$

Other pairwise exponent equalities fail because the remaining term dominates.

13.6 A formal implicit-function principle

A correct dominant balance and invertible linearization usually determine a unique branch, one monomial at a time. In complete transseries settings this becomes a formal implicit-function theorem.

Exercise 13.1. Find five terms of the large solution of $Y^2 - xY - x = 0$.

Exercise 13.2. Solve $Y = x + \log Y$ through x^{-3} .

Exercise 13.3. Use two Newton steps to solve $Y + e^{-Y} = x$ through the e^{-3x} sector.

Lambert's W function as a transseries laboratory

The Lambert function satisfies

$$W(z)e^{W(z)} = z, \quad W(z) + \log W(z) = \log z.$$

Let

$$L_1 = \log z, \quad L_2 = \log L_1.$$

Dominant balance gives $W \sim L_1$, and the next correction is $-L_2$.

14.1 Fixed-point equation for the correction

Write

$$W = L_1 - L_2 + U.$$

Then

$$U = -\log \left(1 - \frac{L_2 - U}{L_1} \right).$$

Seek

$$U = \frac{P_1(L_2)}{L_1} + \frac{P_2(L_2)}{L_1^2} + \frac{P_3(L_2)}{L_1^3} + \dots.$$

Matching coefficients gives

$$\begin{aligned} P_1 &= L_2, \\ P_2 &= \frac{L_2(L_2 - 2)}{2}, \end{aligned}$$

and

$$P_3 = \frac{L_2(2L_2^2 - 9L_2 + 6)}{6}.$$

Thus

$$\begin{aligned} W(z) &= L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2 + L_2)}{2L_1^2} \\ &\quad + \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} \\ &\quad + \frac{L_2(-12 + 36L_2 - 22L_2^2 + 3L_2^3)}{12L_1^4} + \dots \end{aligned}$$

14.2 Residual check

For a truncation W_N , compute

$$R_N = W_N + \log W_N - L_1.$$

Since the derivative with respect to W is $1 + 1/W \sim 1$, the missing correction has the same leading size as $-R_N$.

14.3 Specialization to $W(e^x)$

Setting $L_1 = x$ and $L_2 = \log x$ gives the inverse of $x + \log x$:

$$\begin{aligned} W(e^x) = x - \log x + \frac{\log x}{x} + \frac{(\log x)^2 - 2 \log x}{2x^2} \\ + \frac{2(\log x)^3 - 9(\log x)^2 + 6 \log x}{6x^3} + \dots \end{aligned}$$

14.4 Small argument

Near $z = 0$, the principal branch has the convergent Taylor series

$$W_0(z) = \sum_{n \geq 1} \frac{(-n)^{n-1}}{n!} z^n = z - z^2 + \frac{3}{2} z^3 - \frac{8}{3} z^4 + \dots$$

The real branch $W_{-1}(z)$ tends to $-\infty$ as $z \rightarrow 0^-$. Writing $z = -e^{-x}$ and $W_{-1} = -Y$ gives

$$Y - \log Y = x,$$

which produces a different logarithmic transseries.

14.5 Why this example is universal

Lambert W illustrates how inversion creates the hierarchy

large logarithm $\rightarrow$ logarithm of that logarithm $\rightarrow$ polynomial corrections in their ratio.

Exercise 14.1. Derive the L_1^{-4} coefficient from the fixed-point equation.

Exercise 14.2. Find the first three corrections to the large solution of $Y + 2 \log Y = x$.

Exercise 14.3. Set $z = x^x$ and identify the first four scales in $W(z)$.

Linear differential equations and hidden exponentials

15.1 The Euler equation

For

$$y' + y = \frac{1}{x},$$

seek

$$y_0 = \sum_{n \geq 0} a_n x^{-n-1}.$$

Matching gives $a_0 = 1$ and $a_n = na_{n-1}$, so $a_n = n!$. The homogeneous solution is Ce^{-x} :

$$y = \sum_{n \geq 0} \frac{n!}{x^{n+1}} + Ce^{-x}.$$

Every C gives the same inverse-power expansion.

15.2 Exact solution

An integrating factor gives

$$y = e^{-x} \operatorname{Ei}(x) + Ce^{-x}.$$

Repeated integration by parts yields

$$\operatorname{Ei}(x) \sim \frac{e^x}{x} \left(1 + \frac{1}{x} + \frac{2!}{x^2} + \cdots \right).$$

15.3 Variable coefficient

For

$$y' + \left(1 - \frac{2}{x} \right) y = \frac{1}{x},$$

the homogeneous solution is Cx^2e^{-x} . The perturbative recurrence is

$$a_n = (n + 2)a_{n-1},$$

so

$$y \sim \frac{1}{x} + \frac{3}{x^2} + \frac{12}{x^3} + \frac{60}{x^4} + \cdots + Cx^2e^{-x}.$$

The algebraic prefactor is dictated by the subleading coefficient of the equation.

15.4 Second-order equation

For

$$y'' - y = \frac{1}{x},$$

a particular formal solution is

$$-x^{-1} - 2!x^{-3} - 4!x^{-5} - \cdots.$$

The homogeneous modes are $C_+e^x + C_-e^{-x}$. A condition of at most algebraic growth forces $C_+ = 0$ but leaves C_- undetermined.

15.5 Riccati reduction

For

$$y'' + a(x)y' + b(x)y = 0,$$

put

$$y = \exp\left(\int S dx\right).$$

Then

$$S' + S^2 + aS + b = 0.$$

Dominant balances for S reveal exponential modes; integration of lower corrections produces algebraic prefactors.

15.6 Variation of constants

For

$$y' + p(x)y = q(x),$$

$$y = e^{-P(x)} \left(C + \int^x e^{P(t)} q(t) dt \right), \quad P' = p.$$

The monomial e^{-P} carries the free constant and often matches the optimal-truncation scale of the particular series.

Exercise 15.1. Find the one-parameter transseries solution of $y' + 2y = x^{-2}$ through four algebraic terms.

Exercise 15.2. For $y' + (1 - a/x)y = x^{-1}$, find the recurrence and homogeneous monomial.

Exercise 15.3. Explain why a condition stated only as an inverse-power asymptotic expansion cannot determine the coefficient of e^{-x} .

Nonlinear differential equations and transseries sectors

Nonlinearity multiplies exponentially small corrections, creating a tower of sectors.

16.1 A Riccati model

Study

$$y' + y - y^2 = \frac{1}{x}.$$

For

$$y_0 = \sum_{n \geq 0} a_n x^{-n-1},$$

the recurrence is

$$a_0 = 1, \quad a_m = m a_{m-1} + \sum_{i+j=m-1} a_i a_j.$$

Hence

$$y_0 = \frac{1}{x} + \frac{2}{x^2} + \frac{8}{x^3} + \frac{44}{x^4} + \frac{296}{x^5} + \cdots.$$

16.2 Linearizing around the perturbative sector

Set $y = y_0 + \delta$. Then

$$\delta' + (1 - 2y_0)\delta - \delta^2 = 0.$$

Ignoring δ^2 at first,

$$\frac{\delta'}{\delta} = -1 + 2y_0.$$

Therefore

$$\delta \sim C e^{-x} x^2 \exp\left(-\frac{4}{x} - \frac{8}{x^2} - \frac{88}{3x^3} - \cdots\right).$$

The carrier is $Cx^2 e^{-x}$.

16.3 Sector tower

The nonlinear term $-\delta^2$ has scale $C^2 x^4 e^{-2x}$, forcing a second sector. Repetition leads to

$$y = y_0 + \sum_{k \geq 1} C^k e^{-kx} x^{2k} \phi_k(x), \quad \phi_k \in \mathbb{R}[[x^{-1}]].$$

The free parameter is C ; after normalizing ϕ_1 , the equation recursively determines the sector functions.

16.4 Triangular hierarchy

Let $E = x^2 e^{-x}$. Collecting $C^k E^k$ gives

$$\phi'_k + \left(1 - k + \frac{2k}{x} - 2y_0\right) \phi_k = \sum_{i=1}^{k-1} \phi_i \phi_{k-i}.$$

The right side for sector k depends only on lower sectors.

16.5 Several actions

If linearization produces $e^{-A_1 x}, \dots, e^{-A_r x}$, a nonlinear ansatz is

$$\sum_{\mathbf{k} \in \mathbb{N}^r} C^{\mathbf{k}} e^{-(\mathbf{k} \cdot \mathbf{A})x} x^{\beta(\mathbf{k})} \Phi_{\mathbf{k}}(x^{-1}).$$

Coincident integer combinations of actions create resonance and may force logarithms.

16.6 Formal versus analytic

The sector recursion proves a formal statement. Showing that an actual solution has the transseries requires analytic estimates, integral equations, or summability theorems.

Exercise 16.1. Derive the recurrence for a_m above.

Exercise 16.2. Expand the first-sector prefactor through x^{-3} .

Exercise 16.3. For $y' + y - y^3 = x^{-1}$, determine which exponential sectors can be generated.

WKB expansions and rapidly varying exponentials

17.1 Riccati form

Consider

$$\varepsilon^2 y'' = Q(x)y.$$

Use

$$y = \exp\left(\frac{1}{\varepsilon} \int^x S(t, \varepsilon) dt\right).$$

Then

$$\varepsilon S' + S^2 = Q.$$

With

$$S \sim S_0 + \varepsilon S_1 + \cdots,$$

we obtain

$$S_0 = \pm \sqrt{Q}, \quad S_1 = -\frac{Q'}{4Q}.$$

Thus

$$y_{\pm} \sim Q^{-1/4} \exp\left(\pm \frac{1}{\varepsilon} \int^x \sqrt{Q(t)} dt\right) (1 + \varepsilon a_1 + \cdots).$$

17.2 Airy equation

For $y'' = xy$, the actions are

$$\pm \frac{2}{3} x^{3/2},$$

and the formal modes are

$$x^{-1/4} e^{\pm 2x^{3/2}/3} \left(1 + \sum_{n \geq 1} a_n x^{-3n/2}\right).$$

For the decaying Airy solution,

$$\text{Ai}(x) \sim \frac{1}{2\sqrt{\pi}} x^{-1/4} e^{-2x^{3/2}/3} \left(1 - \frac{5}{48x^{3/2}} + \cdots\right).$$

The differential equation determines the shape; global normalization determines the prefactor $1/(2\sqrt{\pi})$.

17.3 Turning points and Stokes curves

Where $Q(x) = 0$, the WKB branches coalesce and ordinary WKB fails. A rescaled Airy equation is the local model. In the complex plane, curves where exponential phases have equal real parts mark changes in dominance and Stokes switching.

17.4 Large variable and small parameter

With $g = 1/x$,

$$x^\beta e^{-Ax} = g^{-\beta} e^{-A/g}.$$

The two notations describe the same transseries hierarchy.

Exercise 17.1. Apply leading WKB to $y'' = x^2 y$.

Exercise 17.2. For $\varepsilon^2 y'' = (1 + x^2)y$, write the action integral explicitly.

Exercise 17.3. Show that the action difference between the Airy modes is $4x^{3/2}/3$.

Iteration, Abel equations, and fractional iterates

18.1 Continuous interpolation of iterates

A fractional iterate satisfies

$$T^{[s+t]} = T^{[s]} \circ T^{[t]}, \quad T^{[1]} = T.$$

For $T(x) = x + a$, $T^{[s]} = x + sa$. For $T(x) = ax$, $T^{[s]} = a^s x$. For $T(x) = x^a$, $T^{[s]} = x^{a^s}$.

18.2 Abel's equation

If

$$V(T(x)) = V(x) + 1,$$

then

$$T^{[s]} = V^{-1}(V + s).$$

For $T(x) = 2x$, take $V(x) = \log x / \log 2$.

18.3 Near translation

Let

$$T(x) = x + 1 + \frac{a}{x}.$$

Seek

$$V(x) = x - a \log x + \frac{b_1}{x} + \frac{b_2}{x^2} + \cdots.$$

Substitute into $V(T) - V = 1$. Taylor expansion gives

$$b_1 = \frac{a(1-2a)}{2}, \quad b_2 = \frac{a(6a^2-3a+1)}{12}.$$

The logarithm $-a \log x$ records the accumulated a/x drift over repeated steps.

18.4 Iterative logarithm

One may seek a vector field

$$\frac{dx}{ds} = A(x)$$

whose time-one map is T . Formally,

$$T = \exp(A(x)\partial_x), \quad T^{[s]} = \exp(sA(x)\partial_x).$$

The generator A is an iterative logarithm.

18.5 Parabolic dynamics

A map tangent to the identity near zero becomes a near-translation after inversion of the coordinate. Abel functions then naturally involve powers, logarithms, and sometimes exponentially small corrections. This is a classical source of transseries in local dynamics.

18.6 Support subtlety

Composition requires more than termwise substitution. The substituted monomial family must remain summable. Edgar develops the needed recursion and support theory for large positive transseries and fractional iteration [14, 15].

Exercise 18.1. Verify the group law for x^{a^s} .

Exercise 18.2. Check the displayed values of b_1 and b_2 by expanding $V(T) - V$.

Exercise 18.3. Show that replacing an Abel function V by $V + c$ does not change the fractional iterates.

Part IV

From formal symbols to actual functions

Divergence, least terms, and exponentially small remainders

19.1 Asymptotic expansion versus convergence

A function has

$$f(x) \sim \sum_{n \geq 0} a_n x^{-n-1}$$

when every fixed truncation has the prescribed error as $x \rightarrow \infty$. This says nothing about the limit of partial sums for fixed x .

Common trap

The limits $x \rightarrow \infty$ with truncation order fixed and truncation order $N \rightarrow \infty$ with x fixed are different operations.

19.2 Factorial model and least term

For

$$t_n(x) = \frac{n!}{x^{n+1}},$$

$$\frac{t_{n+1}}{t_n} = \frac{n+1}{x}.$$

Terms decrease until $n \approx x$ and then grow. Stirling's formula gives the least-term scale

$$t_x(x) \approx \sqrt{\frac{2\pi}{x}} e^{-x}.$$

Thus optimal truncation predicts the same exponential scale as the homogeneous solution of the Euler equation.

19.3 Optimal truncation

A practical rule is:

1. estimate consecutive-term ratios;
2. truncate near the smallest term;
3. use its size as a first remainder estimate;
4. search for a transseries sector on that exponential scale.

Hyperasymptotics goes further by re-expanding the remainder in adjacent sectors.

19.4 Gevrey order

A series is Gevrey-1 if

$$|a_n| \leq CA^n n!$$

for some $A, C > 0$. More generally, Gevrey- s growth resembles $A^n(n!)^s$. Ordinary Borel summation is naturally adapted to Gevrey-1 series.

If

$$a_n \sim K \frac{\Gamma(n + \beta)}{A^{n+\beta}},$$

then the least term appears near $n \approx Ax$ and has scale e^{-Ax} times a power of x .

19.5 Watson's lemma

If

$$f(x) = \int_0^\infty e^{-xt} g(t) dt, \quad g(t) = \sum_{n \geq 0} c_n t^n$$

near $t = 0$, then under suitable conditions

$$f(x) \sim \sum_{n \geq 0} \frac{c_n n!}{x^{n+1}},$$

because

$$\int_0^\infty e^{-xt} t^n dt = \frac{n!}{x^{n+1}}.$$

This explains the ubiquity of factorial growth in Laplace-type integrals.

19.6 Saddle points

For

$$I(x) = \int_\Gamma e^{-x\phi(t)} a(t) dt,$$

each relevant critical point contributes a sector

$$e^{-x\phi(t_k)} x^{-\beta_k} \sum_{n \geq 0} c_{k,n} x^{-n}.$$

The critical values are exponential actions. Changing the contour or the direction of x can change which saddles contribute.

Big idea

Divergence is often the mechanism by which an algebraic expansion announces hidden exponentially small information.

Exercise 19.1. Estimate the least-term index for $a_n = \Gamma(n + \beta)A^{-n}$ in $a_n x^{-n}$.

Exercise 19.2. Show $e^{-\sqrt{x}}$ is below every power but above e^{-x} .

Exercise 19.3. Explain why coefficients growing like $(2n)!$ require more than the ordinary Borel transform.

Borel transformation and Laplace reconstruction

20.1 Convention

For

$$\tilde{\Phi}(x) = \sum_{n \geq 0} a_n x^{-n-1},$$

define

$$\hat{\Phi}(\zeta) = \mathcal{B}\tilde{\Phi} = \sum_{n \geq 0} \frac{a_n}{n!} \zeta^n.$$

The directional Laplace transform is

$$\mathcal{L}_\theta \hat{\Phi}(x) = \int_0^{e^{i\theta} \infty} e^{-x\zeta} \hat{\Phi}(\zeta) d\zeta.$$

Termwise integration returns $a_n x^{-n-1}$.

20.2 Three hurdles

For Borel summability in a direction, one needs:

1. convergence of $\hat{\Phi}$ near 0;
2. analytic continuation along the Laplace ray;
3. at most exponential growth along that ray.

20.3 Alternating Euler series

For

$$\tilde{F}(x) = \sum_{n \geq 0} \frac{(-1)^n n!}{x^{n+1}},$$

$$\widehat{F}(\zeta) = \frac{1}{1 + \zeta}.$$

There is no positive-axis singularity, so

$$F(x) = \int_0^\infty \frac{e^{-x\zeta}}{1 + \zeta} d\zeta = e^x E_1(x).$$

Differentiation gives

$$F' = F - \frac{1}{x},$$

so the Borel sum solves a linear ODE.

20.4 Products become convolutions

In the Borel plane,

$$\mathcal{B}(\widetilde{F}\widetilde{G}) = \widehat{F} * \widehat{G},$$

where

$$(\widehat{F} * \widehat{G})(\zeta) = \int_0^\zeta \widehat{F}(\eta) \widehat{G}(\zeta - \eta) d\eta.$$

The beta-integral identity

$$\int_0^\zeta \eta^m (\zeta - \eta)^n d\eta = \frac{m!n!}{(m+n+1)!} \zeta^{m+n+1}$$

explains the factorial normalization.

20.5 Differentiation

With our convention,

$$\mathcal{B}(\widetilde{\Phi}') = -\zeta \widehat{\Phi},$$

up to special treatment of polynomial terms. Differential equations become algebraic-convolution equations.

20.6 Euler equation in the Borel plane

For

$$y' + y = \frac{1}{x},$$

the perturbative solution has

$$\widehat{y}(\zeta) = \frac{1}{1 - \zeta}.$$

The singularity at $\zeta = 1$ is the Borel-plane shadow of the exponential e^{-x} .

20.7 Complete sectors

For

$$e^{-kAx} x^{\beta_k} \tilde{\Phi}_k(x),$$

one Borel-sums the fluctuation series $\tilde{\Phi}_k$ and retains the exponential and power prefactors. Summing over k is an additional analytic problem.

20.8 Limitations

A Borel transform may have zero radius of convergence, a natural boundary, excessive growth, or singularities on every useful ray. Generalized transforms and acceleration handle some cases, but no elementary method sums every formal series.

Checkpoint

factorial formal series $\xrightarrow{\mathcal{B}}$ Borel germ
 $\xrightarrow{\text{analytic continuation}}$ Borel-plane function
 $\xrightarrow{\mathcal{L}}$ sectorial analytic function.

Exercise 20.1. Borel sum $\sum (-1)^n n! / (2x)^{n+1}$ as an integral.

Exercise 20.2. Show the Laplace transform of a convolution is the product of Laplace transforms.

Exercise 20.3. Apply the Borel transform to $y' + ay = x^{-1}$.

Stokes phenomenon: when tiny terms switch on

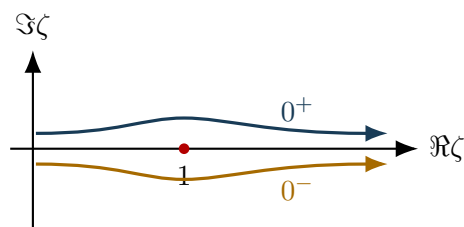
21.1 Lateral sums

For the nonalternating Euler series,

$$\widehat{E}(\zeta) = \frac{1}{1-\zeta}$$

has a pole at $\zeta = 1$ on the positive Laplace ray. Define upper and lower lateral sums by detouring above or below the pole:

$$\mathcal{S}_{0\pm} \widetilde{E} = \int_0^{\infty e^{\pm i0}} \frac{e^{-x\zeta}}{1-\zeta} d\zeta.$$



21.2 Jump

The contour difference gives

$$\boxed{\mathcal{S}_{0+} \widetilde{E} - \mathcal{S}_{0-} \widetilde{E} = 2\pi i e^{-x}.$$

This is a homogeneous solution of $y' + y = x^{-1}$.

21.3 Parameter compensation

For

$$\tilde{y} = \widetilde{E} + C e^{-x},$$

the same analytic function is represented on both sides if

$$C_+ = C_- - 2\pi i.$$

The exact function does not jump; its asymptotic coordinate does.

Big idea

A Stokes jump is a change of asymptotic coordinates. The jump of the perturbative sum is compensated by a jump of the transseries parameter.

21.4 Stokes and anti-Stokes terminology

For exponentials $e^{-S_1/g}$ and $e^{-S_2/g}$, curves where imaginary parts align often support switching, while curves where real parts agree mark equal magnitude. Different communities interchange the labels Stokes and anti-Stokes; always check definitions.

21.5 Smoothing

At finite large parameter, a Stokes multiplier changes rapidly but smoothly, often with an error-function profile. The discontinuous jump is the limiting asymptotic description.

21.6 Median summation

In a simple conjugate situation, the average

$$\mathcal{S}_{\text{med}} = \frac{1}{2}(\mathcal{S}_{0+} + \mathcal{S}_{0-})$$

can produce a real value. Full median resummation in nonlinear resurgence is more structured.

21.7 Stokes matrices

For a linear system, crossing a Stokes ray changes an asymptotic basis by a triangular matrix. The off-diagonal entries are Stokes multipliers; triangularity reflects the hierarchy of dominant and subdominant exponentials.

21.8 Airy example

The Airy actions $\pm 2z^{3/2}/3$ exchange dominance as $\arg z$ changes. The entire function $\text{Ai}(z)$ does not jump, but the efficient combination of exponential asymptotic modes changes sector by sector.

Exercise 21.1. Compute the lateral jump for $1/(A - \zeta)$ with $A > 0$.

Exercise 21.2. If a perturbative jump is $Se^{-Ax}\Phi_1$, find the compensating parameter jump.

Exercise 21.3. Explain why a negative-axis Borel singularity does not obstruct the positive Laplace ray.

Resurgence: sectors talking to one another

22.1 The resurgent feedback loop

A typical transseries is

$$\tilde{\Phi}(x; \sigma) = \sum_{k \geq 0} \sigma^k e^{-kAx} x^{k\beta} \tilde{\Phi}_k(x).$$

Resurgence relates:

- large-order coefficients in one sector;
- Borel singularities;
- low-order coefficients in neighboring sectors;
- Stokes jumps of σ .

22.2 Simple pole and factorial growth

If

$$\hat{\Phi}_0(\zeta) \approx \frac{K}{A - \zeta},$$

then expansion at the origin gives

$$a_{0,n} \approx K \frac{n!}{A^{n+1}}.$$

Distance to the nearest Borel singularity controls late coefficient growth.

22.3 Branch singularities

If

$$\hat{\Phi}_0(\zeta) \sim K(A - \zeta)^{-\alpha},$$

then

$$a_{0,n} \sim \frac{K}{\Gamma(\alpha)} \frac{\Gamma(n + \alpha)}{A^{n+\alpha}}.$$

Subleading local singular terms yield a full inverse- n expansion.

22.4 Neighboring-sector data

In a resurgent problem, the local singularity near A often contains the Borel transform of the next sector, schematically

$$\hat{\Phi}_0(\zeta) \sim \frac{S_{01}}{2\pi i} \hat{\Phi}_1(\zeta - A) \log(\zeta - A) + \cdots.$$

Consequently, late coefficients of Φ_0 encode early coefficients of Φ_1 .

22.5 Numerical extraction

If many $a_{0,n}$ are known, normalize by predicted gamma growth:

$$\frac{A^{n+\beta} a_{0,n}}{\Gamma(n+\beta)}.$$

Sequence acceleration can estimate Stokes constants and neighboring-sector coefficients.

22.6 Alien derivatives

An alien derivative Δ_A measures singular content at Borel location A . A schematic bridge equation is

$$\Delta_A \tilde{\Phi}(x; \sigma) = S_A(\sigma) \frac{\partial}{\partial \sigma} \tilde{\Phi}(x; \sigma).$$

It connects Borel geometry to parameter motion. Precise conventions vary.

22.7 Stokes automorphism

Schematically,

$$\mathfrak{S}_\theta = \exp \left(\sum_{A: \arg A = \theta} e^{-Ax} \Delta_A \right),$$

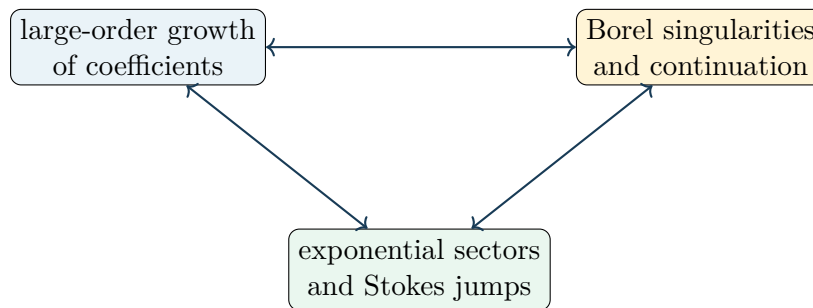
and lateral sums satisfy

$$\mathcal{S}_{\theta+} = \mathcal{S}_{\theta-} \circ \mathfrak{S}_\theta.$$

22.8 Transseries versus resurgence

A formal transseries is defined by ordered support and closure operations. Resurgence is additional analytic continuation structure. The two notions overlap but are not synonymous.

22.9 The triangle



Each corner predicts the others when the resurgent structure is controlled [4, 11].

Exercise 22.1. If $a_n \sim 3n!/2^{n+1}$, predict the nearest simple Borel singularity.

Exercise 22.2. What coefficient pattern can arise from singularities at both A and $-A$?

Exercise 22.3. Explain why factorial growth alone does not prove resurgence.

Multiparameter transseries, resonance, and logarithms

23.1 Sector lattice

With actions A_1, A_2 ,

$$\tilde{\Phi} = \sum_{n,m \geq 0} \sigma_1^n \sigma_2^m e^{-(nA_1+mA_2)x} x^{\beta_{n,m}} \tilde{\Phi}_{n,m}(x).$$

Quadratic nonlinearities add lattice indices.

23.2 Complex actions

Conjugate actions produce decaying oscillations:

$$e^{-(a+ib)x} + e^{-(a-ib)x} = 2e^{-ax} \cos bx.$$

Growing sectors may be excluded by boundary conditions in a chosen direction.

23.3 Resonance

If distinct integer combinations of actions coincide, different sectors occupy the same exponential scale. The recursive equation may become noninvertible and force powers of x or $\log x$.

The elementary equation

$$y'' - y = e^x$$

is resonant: a trial Ae^x fails, while Axe^x works.

23.4 Logarithmic sectors

A resonant hierarchy may require

$$e^{-Ax} x^\beta \sum_{j=0}^J (\log x)^j \Phi_j(x^{-1}).$$

Repeated integration of x^{-1} is the simplest mechanism forcing logarithms.

23.5 Instanton notation

With $g = 1/x$,

$$\Phi(g; \sigma) = \sum_n \sigma^n \exp\left(-\frac{n \cdot A}{g}\right) g^{n \cdot \beta} \Phi_n(g).$$

The algebra is the same sector bookkeeping, though terminology comes from saddle configurations in physics.

23.6 Five levels of validity

For a multiparameter transseries, distinguish:

1. formal coefficient recursion;
2. legal support;
3. Borel summability of each sector;
4. summation over sectors;
5. global matching of parameter values.

Success at one level does not automatically settle the next.

Exercise 23.1. List all two-action sectors with $n + m \leq 3$.

Exercise 23.2. If $A_2 = 2A_1$, identify coincident action labels.

Exercise 23.3. Rewrite $e^{-A/g} g^\beta (\log g)^k$ in $x = 1/g$ notation.

Part V

The surrounding mathematical landscape

Hardy fields and H -fields

24.1 Germs at infinity

Two functions have the same germ at $+\infty$ if they agree for all sufficiently large x . A germ forgets every bounded initial interval and keeps only eventual behavior.

A **Hardy field** is a field of such germs closed under differentiation. Every nonzero member must be eventually nonzero, since its reciprocal must exist. Therefore $\sin x$ cannot belong to a Hardy field: it has infinitely many zeros.

24.2 Eventual order

Define

$$f < g \iff f(x) < g(x) \text{ eventually.}$$

For Hardy-field elements, this is a total order. A nonzero germ has eventual constant sign, and nonconstant elements are eventually monotone after examining their derivatives.

24.3 Formal expansions of germs

Expressions built from x , real constants, exponentials, logarithms, and field operations often belong to Hardy fields and admit corresponding transseries. A faithful expansion map turns eventual comparison into leading-monomial comparison. Not every Hardy field comes equipped with a canonical expansion map, so functions and formal series remain distinct categories.

24.4 The H -field axioms

An H -field is an ordered differential field K abstracting Hardy-field behavior. Let

$$C = \{f : f' = 0\}$$

be the constant field, $\mathcal{O}$ the convex hull of C , and $\mathfrak{o}$ the infinitesimals. A standard formulation includes:

1. if f is larger than every constant, then $f' > 0$;

2. every bounded element is a constant plus an infinitesimal:

$$\mathcal{O} = C + \mathfrak{o}.$$

The transseries field $\mathbb{T}$ is a central example.

24.5 Logarithmic derivatives

The logarithmic derivative

$$f^\dagger = \frac{f'}{f}$$

turns products into sums. For basic monomials,

$$(x^a)^\dagger = \frac{a}{x}, \quad (e^L)^\dagger = L', \quad ((\log x)^b)^\dagger = \frac{b}{x \log x}.$$

Valuations of logarithmic derivatives form an asymptotic couple, an important invariant of an H -field.

24.6 Asymptotic integration and Liouville closure

An asymptotic integral of f is a g with $g' \sim f$. For example,

$$\left(-\frac{e^{-x}}{x}\right)' = \frac{e^{-x}}{x} + \frac{e^{-x}}{x^2} \sim \frac{e^{-x}}{x}.$$

Liouville closure informally means adjoining solutions to

$$y' = f, \quad y'/y = f,$$

so that antiderivatives and exponentials of antiderivatives are present.

24.7 Differential algebra

A differential polynomial is a polynomial in

$$Y, Y', \dots, Y^{(r)}$$

with coefficients in the field. Valuation and differential Newton methods analyze equations

$$P(Y, Y', \dots, Y^{(r)}) = 0.$$

Properties called ω -freeness and newtonianity ensure, roughly, that no asymptotic gaps obstruct natural differential equations.

24.8 Structural results

The monograph of Aschenbrenner, van den Dries, and van der Hoeven develops the ordered differential algebra and model theory of $\mathbb{T}$ [1]. A 2024 theorem by the same authors states that all maximal Hardy fields are elementarily equivalent, as differential fields, to $\mathbb{T}$ [2]. Thus they satisfy the same first-order differential-field statements, even though they are not identified element by element.

24.9 Formal versus functional

	Formal transseries	Hardy-field germs
Elements	symbolic well-based sums	eventual classes of functions
Equality	coefficientwise	eventual equality
Order	leading monomial	eventual pointwise order
Derivative	formal termwise rule	derivative of representatives
Convergence	support summability	analytic behavior

The deepest theory builds bridges without confusing the columns.

Exercise 24.1. Show that rational-function germs form a Hardy field.

Exercise 24.2. Compute the logarithmic derivative of $x^a(\log x)^b e^{-x^c}$.

Exercise 24.3. Give two reasons persistent oscillation is incompatible with a Hardy field.

Surreal numbers and transseries

25.1 Recursive construction

A surreal number is written

$$x = \{L \mid R\},$$

where every member of L is less than every member of R . It is the simplest number lying between them. Early examples are

$$0 = \{\emptyset \mid \emptyset\}, \quad 1 = \{0 \mid \emptyset\}, \quad -1 = \{\emptyset \mid 0\}, \quad \frac{1}{2} = \{0 \mid 1\}.$$

Transfinite continuation creates all reals and many non-Archimedean numbers.

25.2 The infinite number ω

$$\omega = \{0, 1, 2, \dots \mid \emptyset\}$$

is larger than every natural number, and ω^{-1} is a positive infinitesimal. This mirrors

$$x \succ 1 \succ x^{-1}$$

in transseries.

25.3 Conway normal form

Every surreal has a normal form

$$s = \sum_{i < \alpha} r_i \omega^{a_i}$$

with real nonzero coefficients and strictly decreasing surreal exponents. This resembles a Hahn series whose exponent group is itself surreal.

25.4 Exponentials, logarithms, and derivations

The surreal field admits an exponential extending the real exponential. Berarducci and Mantova constructed a natural derivation compatible with this structure and embedded large transseries fields into the surreals [5]. Their later work interprets transseries and larger omega-series as germs of surreal functions [6].

25.5 Evaluation at infinity

A finite transseries can be informally evaluated at $x = \omega$:

$$x + \log x + e^{-x} \mapsto \omega + \log \omega + e^{-\omega}.$$

For an infinite sum, a summability theorem is needed. Under suitable embeddings, this turns formal asymptotic expressions into actual non-Archimedean values.

25.6 What is not identical

The surreal field is far larger than ordinary finite-height, finite-depth LE-transseries. Surreals may have transfinite supports and complexities outside the standard transseries field. A transseries also carries a distinguished variable and composition behavior, while a surreal number is primarily a value.

25.7 Why the bridge matters

The connection:

- links Hahn and Conway normal forms;
- gives non-Archimedean values to asymptotic expressions;
- extends differential algebra to a universal ordered field;
- points toward hyperseries beyond finite logarithmic depth.

Exercise 25.1. Order e^ω , ω^2 , ω , $\log \omega$, 1, and ω^{-1} .

Exercise 25.2. Explain why finite substitution at ω is easier than evaluating an infinite formal sum.

Exercise 25.3. Give a reason the whole surreal field should not simply be called the field of transseries.

How a computer can manipulate transseries

26.1 Recursive representation

A truncated transseries can be stored as an ordered sparse list

$$[(c_0, \mathfrak{m}_0), \dots, (c_N, \mathfrak{m}_N)], \quad \mathfrak{m}_0 \succ \dots \succ \mathfrak{m}_N.$$

A monomial is recursively represented as $x^a e^L$, where L is a purely large transseries object. The core primitive is a comparison routine for monomials.

26.2 Addition

Addition merges ordered lists and combines equal monomials:

```
ADD(T,U):
  compare the next monomial in each list
  copy the larger one to the output
  if equal, add coefficients and discard zero
  continue until both lists are exhausted
```

26.3 Multiplication

For finite truncations, multiply all term pairs, combine equal monomials, sort, and truncate. For infinite supports, Hahn point-finiteness guarantees finitely many contributions to each coefficient. Priority queues can generate products from largest to smallest.

26.4 Reciprocal

Normalize

$$T = c\mathfrak{m}(1 + U), \quad U \prec 1,$$

then generate

$$T^{-1} = c^{-1}\mathbf{m}^{-1}(1 - U + U^2 - \dots)$$

until the next power lies below the cutoff.

26.5 Exponential and logarithm

The software must split $T = L + c + S$ before exponentiating, and split positive $T = c\mathbf{m}(1 + U)$ before taking a logarithm. Without normalization, representations are noncanonical.

26.6 Differentiation

For each term,

$$(c\mathbf{m})' = c\mathbf{m}\mathbf{m}^\dagger.$$

One input term may create several output terms, which must be merged and reordered.

26.7 Composition

A safe composition algorithm must:

1. verify the inner series is large positive;
2. recursively substitute into monomials;
3. expand around dominant parts;
4. prove the substituted family is summable;
5. apply a global cutoff.

Naive termwise truncation can miss contributions lifted above the final cutoff.

26.8 Ratio sets as certificates

Finite ratio sets provide machine-checkable evidence that supports stay in a grid and powers move downward. A trustworthy result should report assumptions, variable direction, cutoff, and whether the output is merely formal or analytically summed.

26.9 A symbolic trace

For $(e^x + x)^{-1}$:

1. leading term e^x ;
2. small remainder $U = xe^{-x}$;

3. ratio set $\{xe^{-x}\}$;
4. powers $U^n = x^n e^{-nx}$;
5. output $e^{-x} - xe^{-2x} + x^2 e^{-3x} - \dots$.

26.10 Limits

Transseries expose cancellation symbolically. For example,

$$\frac{e^{x+e^{-x}} - e^x}{1/x} = xe^x(e^{e^{-x}} - 1) = x \left(1 + \frac{1}{2}e^{-x} + \dots\right),$$

so the expression tends to $+\infty$.

Exercise 26.1. Write pseudocode for truncated multiplication.

Exercise 26.2. Why is monomial comparison needed to compare $x^{100}e^{-x}$ with x^{-1000} ?

Exercise 26.3. Compute $(x + \log x)^{-1}$ through x^{-4} using the reciprocal algorithm.

Which kind of transseries?

27.1 Log-free and logarithmic-exponential

Log-free transseries use power monomials and exponentials without global logarithmic substitution. Full LE-transseries allow finite depth in

$$\log x, \log \log x, \dots$$

Logarithms are essential for integration of $1/x$ and compositional inversion.

27.2 Grid-based and well-based

Grid-based supports lie in finitely generated multiplicative grids and suit algorithms. Well-based supports allow general reverse well-ordered sets and provide broader closure. Edgar's exposition emphasizes real grid-based transseries [13].

27.3 LE and EL constructions

Different staged constructions emphasize logarithmic or exponential closure in different orders. The logarithmic-exponential series of van den Dries, Macintyre, and Marker are central in model theory [10].

27.4 Écalle transseries

Écalle's tradition closely links formal sectors to resurgence, alien calculus, and generalized summation. Analyzable functions are represented through this larger formal-analytic machinery. This overlaps with, but is not identical to, the ordered-field tradition.

27.5 Dulac and power-log series

Singular dynamical systems often produce

$$\sum_{\alpha,k} c_{\alpha,k} x^\alpha (\log x)^k.$$

Dulac transseries add composition and exponentials adapted to return maps and limit-cycle problems.

27.6 Oscillatory transseries

Persistent oscillation does not fit a real total asymptotic order. Complex or oscillatory transseries include

$$e^{-A(x)} e^{i\phi(x)} \quad \text{or} \quad e^{-A(x)} \cos \phi(x),$$

with separate amplitude and phase bookkeeping.

27.7 Multisummability and acceleration

Several Gevrey levels may require repeated Borel transforms, multisummability, or Écalle acceleration. These methods extend analytic reconstruction beyond ordinary Borel summation.

27.8 Hyperseries

Finite logarithmic depth does not capture every natural scale. Logarithmic hyperseries and related constructions use transfinite iteration and connect transseries with maximal Hardy fields and surreal numbers.

27.9 Comparison table

Variant	Typical monomials	Emphasis	Setting
Power/Puiseux	x^q	algebraic branching	algebraic equations
Power-log	$x^a (\log x)^k$	regular singularities	ODEs, dynamics
Grid-based LE	$x^a e^L$ on finite grids	computation	symbolic asymptotics
Well-based LE	Hahn supports	formal closure	differential algebra
Resurgent	$e^{-A/g} g^\beta \Phi(g)$	Borel and Stokes data	ODEs, physics
Oscillatory	$e^{-A} e^{i\phi}$	phase and amplitude	waves, saddles
Hyperseries	transfinite log/exp scales	beyond finite depth	Hardy fields, surreals

27.10 Questions for an unfamiliar paper

Ask:

1. What limit and direction are used?
2. Which monomials are allowed?
3. What support condition is imposed?
4. Is equality formal or after summation?
5. Which Borel convention is used?
6. How are Stokes constants normalized?

Most apparent disagreements are convention mismatches.

Exercise 27.1. Classify $x^{1/3} + x^{-2/3}$, $x^{-1}(\log x)^4$, $e^{-x \log x}$, and $e^{-x} \cos x^2$.

Exercise 27.2. Why does finite logarithmic depth exclude an expression using all iterated logarithms at once?

Exercise 27.3. Why must a Borel convention be fixed before quoting a Stokes constant?

Applications, history, and the big picture

28.1 Special functions and ODEs

Lambert W produces nested logarithms. Airy and Bessel functions exhibit WKB sectors and Stokes switching. Exponential integrals produce factorial divergence. Painlevé equations require nonlinear resurgent transseries.

28.2 Singular perturbation

When a small parameter multiplies the highest derivative, setting it to zero changes the equation's order. Outer series, boundary layers, and exponentially small matching constants naturally form one transseries.

28.3 Dynamical systems

Near nonhyperbolic fixed points, normal forms and return maps involve powers and logarithms. Abel equations and exponentially small separatrix splitting bring in composition and resurgence.

28.4 Computer algebra

Transseries support automated limits, asymptotic comparison, inversion, composition, and formal differential-equation solving. Van der Hoeven develops this computational viewpoint systematically [22].

28.5 Model theory

The field $\mathbb{T}$ is rich yet tame enough for a deep first-order theory. Asymptotic differential algebra plays a role analogous to algebraic closure and real closed fields in ordinary algebra [1].

28.6 Mathematical physics

Perturbation expansions in quantum mechanics, field theory, matrix models, and string theory are often factorially divergent. Resurgent transseries combine perturbative and instanton sectors so ambiguities cancel [4, 11].

28.7 Saddle geometry

Steepest-descent contours decompose into thimbles attached to critical points. Each thimble supplies an exponential sector; Stokes phenomena occur when the thimble decomposition changes.

28.8 Historical timeline

Period	Development
Early 1900s	Hardy studies logarithmic-exponential orders of infinity; Hahn constructs generalized series.
Mid 1900s	Poincaré asymptotics, Borel summation, Stokes analysis, and generalized series mature.
1970s–1990s	Écalle develops transseries, resurgence, alien calculus, and analyzable functions.
1990s–2000s	Exponential fields, LE-series, model theory, and computational transseries are systematized.
2000s–2010s	Connections with H -fields, composition, surreal derivations, and physics expand.
2010s–2020s	Resurgent transseries become widespread; the model theory of $\mathbb{T}$ advances.
2024 onward	Maximal Hardy fields are shown elementarily equivalent to $\mathbb{T}$; work continues on hyperseries and surreal functions.

28.9 Five recurring principles

1. Determine order before coefficients.
2. Factor the dominant term and expand the remainder.
3. Let equations generate new scales.
4. Read divergence as information.
5. Keep formal legality separate from analytic summability.

28.10 Common myths

Just a long series.

A transseries is an ordered algebra with support rules.

Every transseries converges.

Formal summability and numerical convergence differ.

Exponentially small means irrelevant.

Such terms may carry initial data or switching information.

Borel summation removes all ambiguity.

Singular directions create lateral ambiguities.

Resurgence equals transseries.

Resurgence is extra analytic structure.

28.11 Open directions

Research directions include effective hyperseries algorithms, nonlinear summability, embeddings among Hardy fields and surreals, resurgence for PDEs, automated extraction of Stokes data, and links between model-theoretic tameness and analytic continuation.

28.12 How the supplied archive was used

The supplied materials span Edgar's formal grid-based calculus and composition, generalized fields of exponential series, surreal derivations, H -field structure, ODE asymptotics, Borel summability, and resurgence. This tutorial reorganizes those strands into a single learning path and supplements them with current primary-source references.

Exercise 28.1. Explain how one exponential e^{-Ax} can appear from linearization, optimal truncation, a Borel singularity, and a saddle action.

Exercise 28.2. Distinguish Hardy fields, H -fields, surreal numbers, and resurgent functions in one paragraph.

A practical study guide

29.1 Six levels of mastery

1. **Asymptotic literacy:** order mixed monomials and recognize cancellation.
2. **Formal algebra:** normalize, invert, exponentiate, logarithmize, and differentiate.
3. **Support control:** understand reverse well ordering, point-finiteness, and grids.
4. **Equation solving:** use dominant balance, residual correction, and sector ansätze.
5. **Analytic reconstruction:** apply Borel–Laplace transforms and interpret Stokes jumps.
6. **Structural context:** compare H -fields, surreals, computational and resurgent variants.

29.2 Twelve-week route

Week	Topic	Goal
1	Scales	Order mixed monomials.
2	Dominant-factor method	Expand inverses, logs, powers.
3	Supports	Test formal legality.
4	LE construction	Track height and depth.
5	Differential calculus	Derivatives and antiderivatives.
6	Composition	Derive Lambert W .
7	Linear ODEs	Recover hidden exponentials.
8	Nonlinear ODEs/WKB	Build sectors and actions.
9	Divergence	Estimate least terms.
10	Borel–Laplace	Sum Euler series.
11	Resurgence	Relate large order and sectors.
12	Structure	Compare H -fields and surreals.

29.3 Ten questions for every expansion

1. What tends where?

2. What is the monomial order?
3. What is the leading term?
4. Is the support admissible?
5. Which operations produced it?
6. Is equality formal or asymptotic?
7. Does each sector converge or require summation?
8. What are the Stokes directions?
9. What fixes the parameters?
10. What error follows from the cutoff?

29.4 One-page mental model

A common transseries has the nested shape

$$\sum_{\text{sectors}} \underbrace{e^{-\text{action}}}_{\text{beyond all orders}} \underbrace{x^\beta (\log x)^k}_{\text{prefactor}} \underbrace{\sum_n a_n x^{-n}}_{\text{fluctuation}}.$$

Formal theory controls the support and operations. Equations determine sectors and coefficients. Borel analysis turns divergent fluctuations into sectorial functions. Stokes data changes parameter coordinates. H -fields and surreal numbers place the hierarchy in broader algebraic worlds.

Capstone problem set

These problems combine ideas from several chapters. Detailed solutions follow in Appendix B.

Exercise A.1 (Dominance chain). Arrange from largest to smallest:

$$e^{x+\sqrt{x}}, \quad e^x x^{100}, \quad e^x / \log x, \quad e^{x-x^{1/3}}, \quad x^x, \quad e^{x \log \log x}, \quad e^{x^2}.$$

Exercise A.2 (Mixed reciprocal). Expand

$$\frac{1}{e^{2x} + xe^x + \log x}$$

through the e^{-4x} sector.

Exercise A.3 (Nested logarithm). Expand

$$\log(x + \log x + e^{-x})$$

through algebraic order x^{-3} and through $x^{-2}e^{-x}$ in the first exponential sector.

Exercise A.4 (Support test). Which supports are reverse well ordered?

$$\begin{aligned} A &= \{x^{-n} : n \geq 0\}, & B &= \{x^{1/n} : n \geq 1\}, \\ C &= \{e^{-nx} : n \geq 0\}, & D &= \{e^{-x/n} : n \geq 1\}, \\ E &= \{x^n e^{-x} : n \geq 0\}. \end{aligned}$$

Compare formal legality with numerical convergence of the corresponding unit-coefficient sums.

Exercise A.5 (Logarithmic inverse). Solve

$$y + 2 \log y = x$$

through order x^{-3} .

Exercise A.6 (Exponential inverse). Solve

$$Y + e^{-Y} = x$$

through the e^{-4x} sector.

Exercise A.7 (Lambert coefficients). Derive the expansion of $W(z)$ through L_1^{-3} from

$$U = -\log\left(1 - \frac{L_2 - U}{L_1}\right), \quad W = L_1 - L_2 + U.$$

Exercise A.8 (Linear ODE). Find the full one-parameter formal solution of

$$y' + 2y = \frac{1}{x}$$

through five algebraic terms. Identify its first Borel singularity.

Exercise A.9 (Variable-coefficient ODE). For

$$y' + \left(1 - \frac{a}{x}\right)y = \frac{1}{x},$$

find the perturbative recurrence, first four terms, and homogeneous transmonomial.

Exercise A.10 (Nonlinear carrier). For

$$y' + y - y^2 = \frac{1}{x},$$

compute $a_0, \dots, a_4$ in the perturbative sector and derive the first nonperturbative carrier.

Exercise A.11 (WKB). Derive the leading WKB solutions of

$$\varepsilon^2 y'' = (1 + x^2)y$$

and evaluate the action integral.

Exercise A.12 (Alternating Borel sum). Borel sum

$$\sum_{n \geq 0} \frac{(-1)^n n!}{3^{n+1} x^{n+1}}$$

in the positive direction.

Exercise A.13 (Lateral jump). For $A > 0$, compute the upper-minus-lower Laplace jump of

$$\widehat{f}(\zeta) = \frac{1}{A - \zeta}.$$

Exercise A.14 (Large order). If

$$\widehat{\Phi}(\zeta) \sim K(A - \zeta)^{-3/2},$$

estimate the late coefficients of

$$\widetilde{\Phi}(x) = \sum_{n \geq 0} a_n x^{-n-1}.$$

Exercise A.15 (Two Borel singularities). What singularity pattern is suggested by

$$a_n \sim n! A^{-n-1} (K_+ + (-1)^n K_-)?$$

Exercise A.16 (Abel coordinate). For $T(x) = x + 1 + a/x$, verify

$$V = x - a \log x + \frac{a(1 - 2a)}{2x} + \frac{a(6a^2 - 3a + 1)}{12x^2} + O(x^{-3})$$

satisfies $V(T) - V = 1 + O(x^{-4})$.

Exercise A.17 (Hardy-field sign). Prove that a nonzero element of a Hardy field has eventual constant sign.

Exercise A.18 (Surreal normal form). For

$$s = \omega + 3 + 5\omega^{-1} + \omega^{-\omega},$$

identify the leading term, finite part, and largest infinitesimal correction.

Exercise A.19 (Cutoff-aware logarithm). Describe an algorithm for computing $\log T$ through a cutoff when

$$T = c\mathfrak{m}(1 + U), \quad U \prec 1.$$

What certificates are needed for reliable termination?

Exercise A.20 (Formal versus analytic). Give examples of:

1. a formally legal divergent series;
2. a numerically convergent sum whose support is not in a finite grid;
3. two functions with one Poincaré expansion;
4. a series Borel summable in one direction but obstructed in another.

Detailed solutions to the capstone problems

Solution 1: dominance chain

Write every term as an exponential. The relevant exponents are

$$x^2, \quad x \log x, \quad x \log \log x, \quad x + \sqrt{x}, \quad x + 100 \log x, \quad x - \log \log x, \quad x - x^{1/3}.$$

For large x ,

$$x^2 > x \log x > x \log \log x > x + \sqrt{x} > x + 100 \log x > x - \log \log x > x - x^{1/3}.$$

Therefore

$$e^{x^2} \succ x^x \succ e^{x \log \log x} \succ e^{x + \sqrt{x}} \succ e^x x^{100} \succ \frac{e^x}{\log x} \succ e^{x - x^{1/3}}.$$

Solution 2: mixed reciprocal

Factor e^{2x} :

$$\frac{1}{e^{2x} + xe^x + \log x} = e^{-2x}(1 + U)^{-1}, \quad U = xe^{-x} + (\log x)e^{-2x}.$$

Through inner order e^{-2x} ,

$$(1 + U)^{-1} = 1 - xe^{-x} + (x^2 - \log x)e^{-2x} + \dots.$$

Hence

$$e^{-2x} - xe^{-3x} + (x^2 - \log x)e^{-4x} + \dots.$$

Solution 3: nested logarithm

Let $L = \log x$ and factor x :

$$\log(x + L + e^{-x}) = \log x + \log \left(1 + \frac{L}{x} + \frac{e^{-x}}{x} \right).$$

Expanding gives the algebraic part

$$\frac{L}{x} - \frac{L^2}{2x^2} + \frac{L^3}{3x^3} + \cdots.$$

The terms linear in e^{-x} begin

$$\frac{e^{-x}}{x} - \frac{Le^{-x}}{x^2} + \frac{L^2e^{-x}}{x^3} + \cdots.$$

Thus

$$\begin{aligned} \log(x + \log x + e^{-x}) &= \log x + \frac{L}{x} - \frac{L^2}{2x^2} + \frac{L^3}{3x^3} \\ &\quad + \frac{e^{-x}}{x} - \frac{Le^{-x}}{x^2} + \cdots. \end{aligned}$$

Solution 4: support test

A is reverse well ordered: $1 \succ x^{-1} \succ \cdots$. B is also reverse well ordered, because every subset of positive integer indices has a least index and therefore a largest monomial $x^{1/n}$. It is generally not contained in a finite rational-power grid. C is reverse well ordered. D is not: every $e^{-x/n}$ is exceeded by $e^{-x/(n+1)}$, so there is no largest term. E is not: $x^n e^{-x}$ increases with n in the formal order.

For unit coefficients, the A and C sums converge numerically for fixed $x > 1$; the B , D , and E sums diverge because their terms fail to tend to zero. Formal legality and numerical convergence are independent tests.

Solution 5: logarithmic inverse

Let $L = \log x$ and seek

$$y = x - 2L + \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \cdots.$$

Substitute into $y + 2 \log y = x$ and expand $\log y = \log x + \log(y/x)$. Matching powers gives

$$A = 4L, \quad B = 4L^2 - 8L,$$

and

$$C = \frac{16}{3}L^3 - 24L^2 + 16L.$$

Therefore

$$\begin{aligned} y &= x - 2L + \frac{4L}{x} + \frac{4L^2 - 8L}{x^2} \\ &\quad + \frac{\frac{16}{3}L^3 - 24L^2 + 16L}{x^3} + \cdots. \end{aligned}$$

Solution 6: exponential inverse

Put $Y = x + \delta$ and $q = e^{-x}$. Then

$$\delta = -qe^{-\delta}, \quad \delta e^{\delta} = -q.$$

Thus $\delta = W(-q)$. Using

$$W(z) = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + \cdots,$$

we obtain

$$Y = x - q - q^2 - \frac{3}{2}q^3 - \frac{8}{3}q^4 + \cdots.$$

Solution 7: Lambert coefficients

Let $q = L_1^{-1}$ and $L = L_2$, with

$$U = qP_1 + q^2P_2 + q^3P_3 + \cdots.$$

From

$$U = q(L - U) + \frac{q^2}{2}(L - U)^2 + \frac{q^3}{3}(L - U)^3 + \cdots,$$

coefficient matching gives

$$P_1 = L, \quad P_2 = \frac{L^2 - 2L}{2}, \quad P_3 = \frac{2L^3 - 9L^2 + 6L}{6}.$$

Insert these into $W = L_1 - L_2 + U$.

Solution 8: linear ODE

For $y_0 = \sum a_n x^{-n-1}$,

$$2a_0 = 1, \quad 2a_n = na_{n-1}.$$

Hence $a_n = n!/2^{n+1}$ and

$$y = \frac{1}{2x} + \frac{1}{4x^2} + \frac{1}{4x^3} + \frac{3}{8x^4} + \frac{3}{4x^5} + \cdots + Ce^{-2x}.$$

The Borel transform is

$$\widehat{y}(\zeta) = \frac{1}{2 - \zeta},$$

so the first singularity is at $\zeta = 2$.

Solution 9: variable coefficient

Substitution gives

$$a_0 = 1, \quad a_n = (n + a)a_{n-1}.$$

Thus

$$\begin{aligned} y_0 &= x^{-1} + (a+1)x^{-2} + (a+1)(a+2)x^{-3} \\ &\quad + (a+1)(a+2)(a+3)x^{-4} + \cdots. \end{aligned}$$

The homogeneous equation has

$$y_h = C \exp(-x + a \log x) = Cx^a e^{-x}.$$

Solution 10: nonlinear carrier

The recurrence

$$a_m = ma_{m-1} + \sum_{i+j=m-1} a_i a_j$$

gives

$$a_0 = 1, \quad a_1 = 2, \quad a_2 = 8, \quad a_3 = 44, \quad a_4 = 296.$$

Linearization yields

$$\delta' + (1 - 2y_0)\delta = 0,$$

so

$$\delta = C \exp\left(-x + 2 \int y_0 dx\right) \sim Cx^2 e^{-x}.$$

Solution 11: WKB

Here $Q = 1 + x^2$. The leading forms are

$$y_{\pm} \sim (1 + x^2)^{-1/4} \exp\left(\pm \frac{1}{\varepsilon} \int^x \sqrt{1 + t^2} dt\right).$$

Since

$$\int \sqrt{1 + t^2} dt = \frac{1}{2} \left(t\sqrt{1 + t^2} + \operatorname{arsinh} t \right),$$

we obtain

$$y_{\pm} \sim (1 + x^2)^{-1/4} \exp\left[\pm \frac{x\sqrt{1 + x^2} + \operatorname{arsinh} x}{2\varepsilon}\right].$$

Solution 12: alternating Borel sum

The Borel transform is

$$\widehat{F}(\zeta) = \sum_{n \geq 0} \frac{(-1)^n}{3^{n+1}} \zeta^n = \frac{1}{3 + \zeta}.$$

Therefore

$$\mathcal{S}_0 \widetilde{F}(x) = \int_0^\infty \frac{e^{-x\zeta}}{3 + \zeta} d\zeta = e^{3x} E_1(3x).$$

Solution 13: lateral jump

The residue of $e^{-x\zeta}/(A - \zeta)$ at $\zeta = A$ is $-e^{-Ax}$. With the upper-minus-lower convention used here, the contour orientation yields

$$\mathcal{L}_{0+} \widehat{f} - \mathcal{L}_{0-} \widehat{f} = 2\pi i e^{-Ax}.$$

Solution 14: large order

Use

$$(A - \zeta)^{-3/2} = A^{-3/2}(1 - \zeta/A)^{-3/2}.$$

The coefficient formula for $(1 - z)^{-\alpha}$ gives

$$a_n \sim \frac{K}{\Gamma(3/2)} \frac{\Gamma(n + 3/2)}{A^{n+3/2}}.$$

Since $\Gamma(3/2) = \sqrt{\pi}/2$, the prefactor is $2K/\sqrt{\pi}$.

Solution 15: two singularities

A singularity at A produces $n!A^{-n-1}$ growth. A singularity at $-A$ produces an alternating contribution proportional to $(-1)^n n!A^{-n-1}$. Thus the pattern suggests Borel singularities at both $\zeta = A$ and $\zeta = -A$.

Solution 16: Abel coordinate

Let

$$V = x - a \log x + b_1 x^{-1} + b_2 x^{-2}.$$

Taylor-expand $V(x + h)$ with $h = 1 + a/x$:

$$V(x + h) - V(x) = V'h + \frac{1}{2}V''h^2 + \frac{1}{6}V'''h^3 + \dots.$$

The x^{-2} coefficient vanishes when

$$b_1 = \frac{a(1 - 2a)}{2},$$

and the x^{-3} coefficient vanishes when

$$b_2 = \frac{a(6a^2 - 3a + 1)}{12}.$$

The remaining error begins at x^{-4} .

Solution 17: Hardy-field sign

A nonzero Hardy-field germ has a reciprocal, so a representative must be nonzero on some tail. If it changed sign infinitely often on that tail, continuity would force infinitely many zeros. Therefore its sign is eventually constant. Equivalently, a germ with zeros arbitrarily far out cannot be invertible unless it is the zero germ.

Solution 18: surreal normal form

The leading term is ω . The finite part is 3. The infinitesimal tail is

$$5\omega^{-1} + \omega^{-\omega},$$

and its largest term is $5\omega^{-1}$ because $\omega^{-1} \succ \omega^{-\omega}$.

Solution 19: cutoff-aware logarithm

Normalize

$$\log T = \log c + \log \mathfrak{m} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n.$$

Generate powers of U , truncating after every multiplication. Stop when the whole next power lies below the cutoff. The algorithm needs a monomial comparison routine, proof that $U \prec 1$, support or ratio-set data ensuring powers move downward, exact coefficient arithmetic, and a global cutoff adjusted for outer factors.

Solution 20: formal versus analytic

Examples are:

1. $\sum n!x^{-n-1}$ is formally legal but divergent.
2. $\sum 2^{-n}x^{-1/n}$ converges for fixed $x > 1$, but its support is not reverse well ordered, so it is not a legal Hahn transseries (and in particular is not contained in a finite rational-power grid).
3. $f(x)$ and $f(x) + e^{-x}$ have the same inverse-power expansion.
4. $\sum n!x^{-n-1}$ is obstructed along the positive Borel ray by $\zeta = 1$ but is summable along a nonsingular nearby direction.

Quick reference sheets

C.1 Dominance hierarchy

For fixed positive constants and $x \rightarrow +\infty$,

$$e^{e^x} \succ e^{x^2} \succ e^x \succ x^a \succ (\log x)^b \succ 1 \succ (\log x)^{-c} \succ x^{-d} \succ e^{-x} \succ e^{-x^2} \succ e^{-e^x}.$$

To compare exponentials, compare exponents. To compare powers, compare exponents after reducing to a common base.

C.2 Normalization formulas

For nonzero T ,

$$T = c\mathbf{m}(1 + U), \quad U \prec 1.$$

For arbitrary T ,

$$T = L + c + S, \quad L \text{ purely large, } S \prec 1.$$

Then

$$T^{-1} = c^{-1}\mathbf{m}^{-1}(1 - U + U^2 - \dots),$$

$$\log T = \log c + \log \mathbf{m} + U - \frac{U^2}{2} + \dots,$$

$$T^\alpha = c^\alpha \mathbf{m}^\alpha \left(1 + \alpha U + \frac{\alpha(\alpha-1)}{2} U^2 + \dots \right),$$

$$e^T = e^L e^c \left(1 + S + \frac{S^2}{2} + \dots \right).$$

C.3 Derivative dictionary

$$\begin{aligned}
 (x^a)' &= ax^{a-1}, \\
 (\log x)^{b'} &= \frac{b(\log x)^{b-1}}{x}, \\
 (e^L)' &= L'e^L, \\
 (x^a e^L)' &= x^a e^L \left(\frac{a}{x} + L' \right), \\
 (T \circ S)' &= (T' \circ S)S'.
 \end{aligned}$$

C.4 Equation-solving dictionary

Situation	First move
Algebraic equation	Balance at least two dominant terms.
Implicit equation	Use a leading guess and residual correction $-F/F'$.
Fixed point $Y = \Phi(Y)$	Check that differences shrink in asymptotic order.
Linear ODE	Find perturbative recurrence and homogeneous exponentials.
Nonlinear ODE	Linearize around the perturbative sector, then multiply sectors.
Second-order ODE	Use Riccati/WKB to identify actions.
Iteration	Solve Abel's equation or find an iterative logarithm.
Divergent series	Estimate coefficient growth and Borel-transform it.

C.5 Borel–Laplace dictionary

For

$$\begin{aligned}
 \tilde{\Phi}(x) &= \sum_{n \geq 0} a_n x^{-n-1}, \\
 \hat{\Phi}(\zeta) &= \sum_{n \geq 0} \frac{a_n}{n!} \zeta^n, \quad \mathcal{S}_\theta \tilde{\Phi} = \int_0^{e^{i\theta}\infty} e^{-x\zeta} \hat{\Phi}(\zeta) d\zeta.
 \end{aligned}$$

Useful correspondences are

x -plane formal operation	Borel-plane operation
multiplication	convolution
$\frac{d}{dx}$	$-\zeta \times$
factorial growth $n!/A^n$	singularity near $\zeta = A$
sector e^{-Ax}	Borel displacement by A
Stokes jump	lateral contour difference

C.6 Support checklist

Before accepting an infinite formal expression, verify:

1. every nonempty support subset has a largest monomial;
2. each coefficient receives finitely many contributions;
3. infinite families are point-finite;
4. operations preserve the support class;
5. a requested truncation leaves finitely many relevant terms.

C.7 Conversion between large and small variables

With $g = x^{-1}$,

$$x^\beta (\log x)^k e^{-Ax} = g^{-\beta} (-\log g)^k e^{-A/g}.$$

A sector expansion

$$\sum_{n \geq 0} a_n x^{-n}$$

becomes a small- g expansion $\sum a_n g^n$.

Glossary

Action	A quantity A appearing in an exponential scale such as e^{-Ax} or $e^{-A/g}$; often a saddle-value difference or integral of a WKB momentum.
Alien derivative	An operator detecting resurgent singular content at a specified Borel location.
Analyzable function	In Écalle's framework, a function represented through transseries together with generalized summation and continuation procedures.
Anti-Stokes line	Depending on convention, a curve where competing exponentials have equal magnitude; terminology varies.
Asymptotic couple	An algebraic structure recording the interaction between valuation and logarithmic differentiation in an H -field.
Asymptotic expansion	A sequence of terms whose every fixed truncation approximates a function to the scale of the next term.
Asymptotic scale	A sequence $\phi_{n+1} \prec \phi_n$ of progressively smaller comparison functions.
Beyond all orders	Smaller than every term of a chosen scale, such as $e^{-x} \prec x^{-N}$ for every N .
Borel plane	The ζ -plane in which factorially normalized coefficients define the Borel transform.
Borel summation	Reconstruction of a function by Borel transformation, analytic continuation, and Laplace integration.
Composition	Substitution of one transseries into another, under largeness and support conditions.
Conway normal form	A generalized series representation $\sum r_i \omega^{a_i}$ of a surreal number.
Cutoff monomial	The smallest magnitude retained in a truncated formal calculation.
Differential polynomial	A polynomial in $Y, Y', \dots, Y^{(r)}$ with coefficients in a differential field.

Dominant balance

A selection of two or more largest terms in an equation that can cancel at leading order.

Dulac series

Power-log or transseries expansions adapted to singular dynamical systems and return maps.

Écalles acceleration

A generalized summation change of scale used when ordinary Borel–Laplace summation is inadequate.

Exponential height

A measure of the number of nested exponential construction stages needed for a transseries.

Flat function

A function invisible to a chosen asymptotic scale, such as e^{-x} relative to inverse powers.

Formal convergence

Stabilization of coefficients above every fixed monomial cutoff, not numerical convergence.

Formal transseries parameter

A constant symbol such as C or σ labeling solution families and exponential sectors.

Germ at infinity

An equivalence class of functions that agree eventually.

Gevrey order

A measure of coefficient growth, with Gevrey-1 corresponding roughly to $n!$ growth.

Grid

A finitely generated multiplicative set of monomials built from small ratios.

Hahn field

A field of generalized series with coefficients in a field and well-ordered support in an ordered abelian group.

Hardy field

A differential field of germs at infinity.

 H -field

An ordered differential field satisfying axioms abstracting eventual growth and bounded decomposition in Hardy fields.

Homogeneous sector

A free solution of the linearized or homogeneous equation, often carrying an exponential monomial.

Hyperasymptotics

Re-expansion of optimally truncated remainders using neighboring exponential sectors.

Hyperseries

Generalized transseries permitting scales beyond finite iterated logarithmic-exponential depth.

Infinitesimal

An element smaller in magnitude than every positive real constant.

Instanton sector

Physics terminology for a sector multiplied by $e^{-A/g}$ or its integer powers.

Iterative logarithm

A formal vector field whose time-one map is a given composition map.

Lambert W

The inverse of $w \mapsto we^w$, a standard source of nested-logarithm transseries.

Lateral Borel sum

A Laplace integral taken just above or below a singular ray.

Leading coefficient

The coefficient of the largest monomial in a nonzero transseries.

Leading monomial

The largest support monomial of a nonzero transseries.

LE-series

Logarithmic-exponential generalized series; details depend on the construction.

Liouville closed

Roughly, closed under antiderivatives and exponential integration needed for first-order linear equations.

Logarithmic depth

A measure of the number of outer iterated logarithmic substitutions required.

Logarithmic derivative

$$f^\dagger = f'/f.$$

Magnitude

The leading monomial of a transseries, ignoring its nonzero coefficient.

Median summation

A symmetry-preserving combination of lateral sums, generalized through the Stokes automorphism.

Monomial group

The ordered multiplicative group of pure asymptotic scales.

Multisummability

Summability involving several Gevrey levels or successive Borel–Laplace procedures.

Newton polygon

A geometric device that predicts valuations or leading powers of solutions from competing terms.

Newtonian H -field

An H -field satisfying a differential analogue of henselian solvability for Newton problems.

Non-Archimedean field

An ordered field containing infinite and infinitesimal elements.

Normal form

A canonical decomposition separating monomial, coefficient, and small unit.

Optimal truncation

Stopping a divergent asymptotic series near its least term.

Point-finite family

An indexed family in which every monomial appears in only finitely many members.

Purely large transseries

A transseries all of whose monomials exceed 1.

Ratio set

A finite set of small monomials generating or witnessing a grid.

Resonance

Coincidence of sector actions or a forcing scale with a homogeneous scale, often producing powers or logarithms.

Resurgence

Controlled Borel analytic continuation in which singularities encode other sectors and large-order behavior.

Reverse well ordered

Every nonempty subset has a largest element in the dominance order.

Sector

A block of terms sharing an exponential action, such as $e^{-kAx} x^\beta \Phi_k(x^{-1})$.

Stokes automorphism

The formal automorphism encoding cumulative jumps across a singular direction.

Stokes constant

A normalization-dependent constant connecting Borel singularities, sector jumps, and large-order growth.

Stokes phenomenon

Direction-dependent change in an asymptotic representation or summation across special rays.

Support

The set of monomials with nonzero coefficients.

Summable family

A family with admissible union support and finite contribution to every coefficient.

Surreal number

An element of Conway's universal ordered field built recursively from left and right sets.

Transasymptotic expansion

An expansion that reorganizes or sums across exponentially indexed sectors.

Transmonomial

A pure monomial scale involving powers, logarithms, exponentials, or their iterations.

Transseries

An asymptotically ordered generalized series with a rich monomial group and specified closure operations.

Turning point

A point where WKB characteristic roots merge, often modeled locally by the Airy equation.

Valuation

An additive measure of smallness; larger valuation usually means smaller magnitude.

Well-based transseries

A transseries with general well-founded support rather than a finitely generated grid support.

Witness

Finite ratio-set data certifying a comparison, support property, or convergence statement.

WKB expansion	A rapidly varying exponential ansatz with an action and a formal fluctuation series.
ω -free	A technical H -field property excluding certain asymptotic gaps and central to the model theory of transseries.

Further reading guide

E.1 Best first formal source

Edgar’s “Transseries for Beginners” is a concise non-specialist entry point to real grid-based transseries. It emphasizes the ordered field, monomials, grids, differentiation, integration, and examples [13]. His companion papers treat ratios, composition, convergence, and fractional iteration [16, 14, 15].

E.2 Systematic computational theory

Van der Hoeven’s *Transseries and Real Differential Algebra* develops transseries as an effective computational and differential-algebraic system [22].

E.3 Model theory and H -fields

The book by Aschenbrenner, van den Dries, and van der Hoeven is the comprehensive structural reference [1]. Their expository ICM article provides a shorter overview, and the maximal-Hardy-field preprint records a major recent theorem [2].

E.4 Surreal connections

Berarducci and Mantova’s papers establish a compatible surreal derivation and interpret transseries as germs of surreal functions [5, 6].

E.5 Borel summation and resurgence

Costin’s book gives a rigorous introduction to asymptotics and Borel summability. Dorigoni offers a compact introduction to resurgence and alien calculus, while Aniceto, Başar, and Schiappa provide a very extensive physics-oriented primer with explicit large-order examples [11, 4].

E.6 How to progress

A productive sequence is:

1. work through Edgar while reproducing every expansion;
2. learn Hahn-series support independently of analysis;
3. solve linear and nonlinear ODE examples by hand;
4. study Borel summation on Euler and Airy models;
5. only then approach alien calculus or advanced H -field model theory.

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